

Erdős Problem #647: A Lean-Verified Structural Reduction and a Conditional Dickson Dichotomy

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Abstract

Erdős Problem #647 asks whether any $n > 24$ satisfies $\max_{m < n} (m + \tau(m)) \leq n + 2$. Building on earlier reductions, we show that any solution has the form $n = 2520N$ with N in one of 96 residue classes modulo $M = 46189 = 11 \cdot 13 \cdot 17 \cdot 19$. Under exact-v2 closure semantics (which retires the prior peeled-cofactor bridge after it was shown unsound), we Lean 4-verify 55 of these classes via explicit closure certificates (42 single-overlap branch contradictions and 13 direct full-value arguments). The remaining 41 unresolved residues decompose as the historical hard-5 structural core $\{0, 1716, 4862, 16302, 17160\}$, nine additional hard residues $\{2431, 9009, 9867, 17017, 21164, 24453, 29315, 31603, 38896\}$ reopened by the exact-v2 recertification audit, and 27 residues whose closures under the Bridge-3 peeled-cofactor mechanism were invalidated by the exact-v2 audit (and remain formally open pending a sound replacement). The entire formal chain is stated conditional on a single intentional axiom `openResiduesStage1_no_solution` asserting that no N with $N \bmod M$ in the open set yields a #647 candidate.

On each of the historical hard-5 classes, the divisor constraints reduce the problem to simultaneous primality of an explicit 13-, 15-, or 17-term linear tuple in a parameter t : a mod-11 square obstruction upgrades the budget- ≤ 3 forms to primes, and residue-specific divisibility by 17 or 19 yields additional prime-quotient constraints. We prove a completeness theorem: at the budget bound $B_k \leq 5$, the resulting list of forced prime-or-prime-quotient conditions consists of exactly one universal extension and six residue-specific triples, and this bound is sharp.

We then prove that the operative local congruence program is exhausted: no higher-power obstruction arises at 17 or 19, and non-operative primes act only by residue translation. We introduce the *offset atlas*, an exact per-prime branch classification of the affine forms $F_{r,k}(t) = 2520(Mt + r) - k$; the atlas unifies the mod-11 square-kill and the residue-specific extensions under a single framework and provides machine-readable data against which the divisor-sum conditions at every integer offset k can be tested. An empirical investigation of the Erdős-inequality-failure pattern yields *killer sets*: small sets $S_r \subseteq [1, 100]$ such that every observed strong near-miss at residue r fails the Erdős inequality at some $k \in S_r$. Freezing the empirical claims to the audited 301-record strong $r = 0$ corpus (and the combined 303-record strong corpus across $r \in \{0, 1716, 4862\}$), the size-3 set $\{15, 16, 27\}$ is the unique minimum cover at $r = 0$ (301/301), while the empirical monitor $\{7, 11, 13, 16\}$ covers the audited strong corpus completely. A confirmed 13-prime hit at $t = 11,741,230,040,310$ is consistent with Bateman–Horn’s leading-order quantitative prediction at the current search range.

We further identify the structural mechanism governing the empirical obstruction: on each atlas branch, the Erdős inequality at offset k reduces to a divisor-count constraint $\tau(Z_{r,k,\lambda}) \leq B_{r,k,\lambda}^{\text{eff}}$ on the primitive cofactor, where the branchwise effective budget $B_{r,k,\lambda}^{\text{eff}} = \lfloor (k+2)/\tau(A_{r,k,\lambda}) \rfloor$ is computable from the atlas alone. The branchwise severity ordering induced by this criterion

matches empirical pass-rate ordering across all 87 offsets $k \leq 100$ at $r = 0$ with Spearman correlation 0.93–0.94 under both empirical and neutral branch weights, stable under chronological splitting ($\rho = 0.96$ between halves) and, for the neutral-prediction audit, bootstrap resampling (median $\rho = 0.93$, 95% CI [0.92, 0.94]). The monitor set $\{7, 11, 13, 16\}$ has an explicit structural interpretation: $k = 16$ is atlas-tight at every hard residue, while $k = 7, 11, 13$ provide residue-specific tightness at the residues where their primes are in the atlas active set. A negative structural result is also reported: three tested finite-incompatibility programs (covering at small primes, pattern-rigidity, pairwise gcd identities) do not produce a contradiction in the killer-set system at $r = 0$. The remaining gap is analytic: a density-level control on the budget- ≥ 4 tail along Dickson sets. The nearest literature-shaped proxy target is a tuple-restricted Titchmarsh-type divisor bound on Dickson sets. At the reference hardest residue $r = 0$ (tuple rank 13) we establish the *absolute form* of this proxy (Remark 11.3): $\sum_{t \leq X, t \in D_0(X)} \tau(L_{0,k_0}(t)) \ll X (\log X)^{-12}$ via a Selberg Λ^2 upper-bound sieve combined with Shiu/Nair–Tenenbaum Brun–Titchmarsh. This absolute bound is unconditional. The bridge from the absolute form to a density-level statement inside the rank-13 Dickson set sits inside the same Dickson/Schinzel-Hypothesis-H conjectural frame that Proposition 11.2 already uses to state the Track-2 dichotomy; it is flagged in Remark 11.3 as a conditional outlook, not as a theorem of this paper. The analogous statements at the other four hard residues are expected to hold by the same composite but require residue-specific bookkeeping (rank 15 or 17 sieve integrals, per-residue admissibility, per-residue resultant support) which is not executed here. A separate Dirichlet- L pair-correlation route is secondary rather than load-bearing here: in the current project notes it gives a conditional asymptotic theorem under GRH plus a χ -twisted Hardy–Littlewood hypothesis H_χ , with a finite- T^* numerical corollary under an additional bookkeeping gate B_χ ; the strict GRH-only finite- T^* individual- χ version is effectively red in the present framework. Thus the remaining content is reduced to five explicit admissibility problems whose small-offset obstruction profile is now structurally characterized by the atlas’s branchwise effective budget distribution.

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1 Problem statement

Let $\tau(m)$ denote the number of positive divisors of m .

Erdős Problem #647. Does there exist $n > 24$ such that

$$\max_{1 \leq m < n} (m + \tau(m)) \leq n + 2?$$

Equivalently, $\tau(n - k) \leq k + 2$ for all $1 \leq k < n$. The known solutions for $n \leq 24$ are $\{1, 2, 3, 4, 5, 6, 8, 10, 12, 24\}$; none exist below 10^{10} (OEIS A087280). The problem remains open.

Erdős conjectured the stronger statement $\max_{m < n} (\tau(m) + m - n) \rightarrow \infty$ (recorded as `variants.lim` in the DeepMind formal conjectures repository [5]), which would imply at most finitely many solutions.

A recent submission [8] proposed a negative resolution via a record-holder domination framework; the unresolved step is the asymptotic claim that domination intervals eventually cover all sufficiently large n . The companion Lean artifact axiomatized that asymptotic step rather than proving it, a structural gap that Tao [9] flagged as yielding “no certificate of unconditional correctness.” Our development avoids this failure mode in a specific sense: the Lean chain contains a single intentional axiom, `openResiduesStage1_no_solution`, whose scope is exactly the 41 residues in the explicit finite set `openResiduesStage1` (defined as the complement of the 55 Lean-closed residues inside the 96-class sieve frontier; see Section 14). That scope is computable and enumerable; it is not a blanket “all sufficiently large n ” axiom. For four of the five historical hard residues $\{1716, 4862, 16302, 17160\}$ (a subset of `openResiduesStage1`), a $[1, 10^{15})$ exhaustive parity scan at $t \equiv r \pmod{M}$ provides empirical stopping certificates recording 0 full-tuple witnesses.¹ We do not claim these stopping

¹Search methodology, summarized from `analysis/k12_k13_structure/findings/hard5_stopping_certificates_20260423.md`: per-residue candidate iteration under `-restrict-modM` (only $t \equiv r \pmod{M}$), Eratosthenes sieve to limit 10^3 with primes dividing M auto-skipped, BPSW primality pre-test on sieve-survivors, then per-form deterministic Miller–Rabin with 12 bases (valid up to 3.3×10^{24}) on each $\alpha_i t + \beta_i$. Each residue processed $\approx 2.17 \times 10^{10}$ candidate t -values; combined across the four residues the campaign primality-tested on the order of 10^5 sieve-survivors.

certificates are proofs; they bound where counterexamples within those four classes could live and are reported as empirical evidence supporting the axiom’s scope.

Two elementary forum derivations (Dutta [6] on 18 Jan 2026 and Alexeev [7] on 19 Jan 2026) independently arrive at the same $n = 2520N$ structure and the same tuple of forms $d_i N - 1$ that our Section 3 derives in full (modulo the subsequent refinements $315N - 1 \in \{p, 2p\}$ and $105N - 1 \in \{p, p^2, 2p, 4p\}$ from Dutta’s 19 Jan follow-up). Our $n = 2520N$ reduction and the forced-prime conditions at $k = 1, \dots$ are consistent with theirs, and our further reduction to 96 classes modulo M can be read as a formalization of Alexeev’s conjecture $11 \mid n, 13 \mid n$ into the explicit mod-46189 frame. The *explicit near-witness* $n = 604,517,614,941,240$ ($N = 239,887,942,437$) exhibited by Alexeev satisfies $\tau(n - k) \leq k + 2$ for $1 \leq k \leq 13$ and $k = 24$, demonstrating that the pointwise tuple conditions alone are not contradictory; the analytic content of #647 is concentrated in the uniformity across all shifts $k < n$.

Our contribution is a structural reduction that localizes any failure of asymptotic coverage to the 41 residues (decomposing as the historical hard-14 subset plus 27 residues whose prior closures were invalidated by the exact-v2 audit’s correction of the Bridge-3 peeled-cofactor mechanism). Section 5 analyses the historical hard-5 structural core in detail; Sections 13–14 record the formal chain.

2 Verified reduction

- (R1) Every solution $n > 84$ has $n = 2520N$ (Lean-verified: `erdos647_div_2520`).
- (R2) Twelve shape constraints reduce $N \bmod 46189$ to exactly 96 residue classes (Lean-verified: `surviving_card`).
- (R3) Under exact-v2 closure semantics, adopted after the pointwise-unsound Bridge-3 peeled-cofactor transfer was retired (see Section 14 and the post-retirement audit at `docs/audits/55_closed_post_bridge3_audit_20260424.md`), 55 of 96 classes are eliminated by Lean-checked closure certificates operating directly on the full value $m = 2520 \cdot N - k$:
 - 42 residues by *single-overlap* arguments combining divisor monotonicity $\tau(d) \leq \tau(m)$ when $d \mid m$ with sound coprime-multiplicativity $\tau(m) \geq 2\tau(a)$ under an explicit hypothesis $\gcd(p, a) = 1$ (Lean-verified: `singleOverlapResidues42`);
 - 13 residues by *direct full-value* closure, the same full-value toolkit specialised to fixed content $g_{\text{exact}} \in \{1729, 2584, 4522, 4845, 9690\}$ (Lean-verified: `directFullValueResidues13`).

The remaining 41 residues form the proof-critical open set `openResiduesStage1 = survivingResidues96 \ closedResiduesStage1`.

- (R4) The 41-residue open set decomposes as:
 - the historical hard-14 core
$$\{0, 1716, 2431, 4862, 9009, 9867, 16302, 17017, 17160, 21164, 24453, 29315, 31603, 38896\}$$
(the original hard-5 structural subset $\{0, 1716, 4862, 16302, 17160\}$ plus nine residues reopened by the exact-v2 recertification audit);
 - the 27 residues whose closures under the now-invalidated Bridge-3 peeled-cofactor mechanism require a sound replacement.

The Stage-1 axiom `openResiduesStage1_no_solution` asserts that no N with $N \bmod M$ in this 41-residue set yields a #647 candidate.

3 The 12-shape almost-prime package

Proposition 3.1 (Extended shape package). *If $n = 2520N$ satisfies the hypothesis, then twelve linear forms $dN - 1$ ($d \in \{105, 140, 210, 252, 280, 315, 420, 504, 630, 840, 1260, 2520\}$) satisfy explicit almost-prime shape conditions with bounded exponents.*

The six pure-prime shapes ($k = 1, 2, 3, 4, 6, 12$) and six almost-prime shapes ($k = 5, 8, 9, 10, 18, 24$) are derived by case analysis of $\tau(2520N - k) \leq k + 2$. Shape elimination promotes $k = 20$ from budget-3 to budget-2, giving 13 prime forms at $r = 0$. The full derivation is formalized in `Erdos647BridgeV1.lean` (863 lines).

At each hard residue $r \in \{0, 1716, 4862, 16302, 17160\}$, the budget- ≤ 3 prime tuple contains a universal 13-form core indexed by $k \in \{1, 2, 3, 4, 5, 6, 8, 9, 10, 12, 18, 20, 24\}$ with coefficients $\alpha_i = d_i \cdot 46189$ for $d_i \in \{2520, 1260, 840, 630, 504, 420, 315, 280, 252, 210, 140, 126, 105\}$. Residue-specific extras extend the full tuple to sizes 13, 17, 15, 15, 17 for $r \in \{0, 1716, 4862, 16302, 17160\}$ respectively; these extras arise when $\gcd(\alpha_k, \beta_k^{(r)})$ picks up residue-dependent factors of the sieve primes $\{11, 13, 17, 19\}$ absent from the core. See `data/hard_root_forms_r*.json` for the per-residue breakdown.

4 Closure mechanisms

4.1 Budget- ≤ 1 contradiction

For 72 residues, some k -value produces a form whose total content $g = \gcd(2520 \cdot 46189, |2520r - k|)$ satisfies $\tau(g) > k + 2$, making the constraint $\tau(2520N - k) \leq k + 2$ impossible.

4.2 Budget-2 inadmissibility

For 5 residues, the set of budget-2 forms (those requiring $\tau(\text{quotient}) \leq 2$, i.e., primality) covers all of $\mathbb{Z}/p\mathbb{Z}$ at some prime p . Every t has at least one budget-2 form divisible by p , hence composite.

4.3 Budget-2/3 inadmissibility (b23 mechanism)

Theorem 4.1 (b23 branch-closure, Lean-verified). *Let $\{L_i(u) = a_i u + b_i\}$ be affine forms with budgets 2 or 3. If a prime p covers all residues mod p via these forms, and every covering form satisfies $L_i(u) > p^2$ for $u \geq U$, then no $u \geq U$ satisfies all budget constraints simultaneously.*

Proof. Fix $u \geq U$. Some covering form L_i has $p \mid L_i(u)$. If budget = 2: $L_i(u) > p$ forces $\tau(L_i(u)) \geq 3 > 2$. If budget = 3: $L_i(u) > p^2$ forces $\tau(L_i(u)) \geq 4 > 3$ (since $1, p, L_i(u)/p, L_i(u)$ are four distinct divisors when $L_i(u)/p > p$). Either contradicts the budget. $\square$

For 10 residues, the b23 mechanism applies at the t -level (without branching). For 4 residues ($r \in \{858, 2574, 24310, 25168\}$), branching $t = 30u + c$ followed by b23 at the u -level closes all 30 branches.

5 Structural characterization of the hard set

Proposition 5.1 (Necessary condition, Lean-verified). *Every hard residue satisfies $143 \mid r$, where $143 = 11 \cdot 13$.*

This has a clean structural consequence: since $11 \cdot 13 \mid r$ and 7 divides every d -coefficient, the forms $\alpha_i = d_i \cdot 46189$ satisfy $\alpha_i \equiv 0 \pmod{q}$ for $q \in \{7, 11, 13\}$. Hence these three sieve primes contribute zero forbidden residues, nullifying three of four operative sieve primes. Only $\{2, 3, 5, 17, 19\}$ remain.

Remark 5.2. The condition $143 \mid r$ is necessary but not sufficient: 43 of 48 survivors with $143 \mid r$ close via budget or inadmissibility mechanisms at the remaining primes. The 5 hard residues are those where even primes $\{2, 3, 5\}$ fail to produce a covering or budget contradiction.

Remark 5.3 (Sieve affinity for 143-divisibility). Of the 96 surviving residues mod 46189, exactly 48 (that is, 50%) are divisible by 143. In the ambient space mod 46189, only $323/46189 \approx 0.70\%$ of residues are 143-divisible. The 12-shape sieve thus exhibits a $71.5\times$ affinity for 143-divisible residues, consistent with the observation that 143-divisibility nullifies the sieve's most aggressive primes (11 and 13). Hardness, once 143-divisibility is present, is then decided among these 48 residues by the behaviour at the residual operative primes $\{2, 3, 5, 17, 19\}$.

Proposition 5.4 (Operative-prime staticity at 17 and 19). *For every shape k in the 13-form core tuple and every residue r mod 46189, the leading coefficient $\alpha_k = d_k \cdot 46189$ satisfies $\alpha_k \equiv 0 \pmod{17}$ and $\alpha_k \equiv 0 \pmod{19}$. Consequently*

$$L_{r,k}(t) \equiv \beta_k^{(r)} \pmod{17} \quad \text{and} \quad L_{r,k}(t) \equiv \beta_k^{(r)} \pmod{19}$$

for every $t \in \mathbb{Z}$: divisibility of $L_{r,k}(t)$ by 17 or 19 is determined entirely by the constant term $\beta_k^{(r)}$, independently of the t -parameter.

Since $46189 = 11 \cdot 13 \cdot 17 \cdot 19$ and every core d -coefficient is a multiple of 7, the factorisation $\alpha_k = d_k \cdot 46189$ forces $\alpha_k \equiv 0$ modulo each of $\{7, 17, 19\}$ as a structural feature of the reduction itself, independent of r . The 143-divisibility condition of the necessary-condition proposition adds $\{11, 13\}$ to this list whenever $143 \mid r$. Among the five operative sieve primes $\{2, 3, 5, 17, 19\}$ that remain after nullifying $\{7, 11, 13\}$, Proposition 5.4 thus isolates $\{17, 19\}$ as carrying no t -parameter information: their contribution to closure or hardness is determined *rigidly* by the constants $\beta_k^{(r)} \pmod{17}$ and $\beta_k^{(r)} \pmod{19}$. This is the structural content that the CRT factorization (Proposition 5.5) observes.

Proposition 5.5 (CRT factorization of hardness, Lean-verified). *Let $\pi: \mathbb{Z}/46189\mathbb{Z} \rightarrow (\mathbb{Z}/17\mathbb{Z}) \times (\mathbb{Z}/19\mathbb{Z})$ be the natural CRT projection. Restricted to the 48 surviving residues with $143 \mid r$, the projection π is injective on both the hard and closing subsets, and their images are disjoint:*

$$\pi(\text{hardResidues5}) = \mathcal{H}_{17,19} := \{(0, 0), (16, 6), (0, 17), (16, 0), (7, 3)\},$$

$$\pi(\text{closingResidues}) \cap \mathcal{H}_{17,19} = \emptyset.$$

Consequently, for $r \in \text{survivingResidues}$,

$$r \in \text{hardResidues5} \iff 143 \mid r \text{ and } \pi(r) \in \mathcal{H}_{17,19}.$$

(Lean theorem `hard_residue_crt_factorization, native_decide.`)

The content of Proposition 5.5 is the injectivity and disjointness, not the enumeration of $\mathcal{H}_{17,19}$: the statement is that the $(\text{mod } 17, \text{mod } 19)$ projection is sufficient statistics for hardness, given 143-divisibility. Any future structural characterization of the historical hard-5 subset within the original 91/5 partition must factor through this projection. The hardness of the five residues therefore appears to arise from the simultaneous failure of all five operative closure mechanisms at $\{2, 3, 5, 17, 19\}$, rather than from a unifying structural property. Whether $\mathcal{H}_{17,19}$ admits a deeper arithmetic characterization is an open question we flag for the community.

Remark 5.6 (Axis overrepresentation in $\mathcal{H}_{17,19}$). Within $\mathcal{H}_{17,19}$, three of the five pairs lie on a coordinate axis — $(0, 0)$, $(0, 17)$, and $(16, 0)$ — compared to only $17 + 19 - 1 = 35$ of $17 \cdot 19 = 323$ pairs (roughly 10.8%) in the ambient space. The hard set is therefore about $5.5\times$ overrepresented on the axes. Arithmetically, this says three of the five hard residues have an *additional* divisibility by one of the residual sieve primes 17 or 19 on top of 143-divisibility: $r = 0$ is divisible by both 17 and 19, $r = 4862$ is divisible by 17, and $r = 16302$ is divisible by 19. The remaining two hard residues, $r = 1716$ and $r = 17160$, have neither additional divisibility but nonetheless project into $\mathcal{H}_{17,19}$ for reasons we have not characterized structurally.

Remark 5.7 (Jacobi-sign necessary condition). Complementing the axis pattern, a cross-Jacobi observation: for each of the 48 surviving residues r with $143 \mid r$, form the pair of Jacobi symbols $\left(\frac{r}{17}, \frac{r}{19}\right)$ and their product $\left(\frac{r}{17}\right) \cdot \left(\frac{r}{19}\right) \in \{-1, 0, +1\}$. Of the 48 surviving 143-divisible residues, exactly 18 have product -1 (the “mixed QR/NR” case, where one of $r \bmod 17, r \bmod 19$ is a quadratic residue and the other is a non-residue), and *zero* of those 18 are hard; the five hard residues all have product in $\{0, +1\}$. Explicitly:

$$\left(\frac{r}{17}\right) \cdot \left(\frac{r}{19}\right) = \begin{cases} 0, & r \in \{0, 4862, 16302\}, \\ +1, & r \in \{1716, 17160\}. \end{cases}$$

Hence a partial necessary condition: *if $r \in \text{survivingResidues}$, $143 \mid r$, and the Jacobi product equals -1 , then r is closing*. The condition is not sufficient — 15 of the 30 surviving 143-divisible residues with non-mixed Jacobi product are also closing — but it strictly reduces the search space for hardness by 60% (from 48 to 30 candidates). The observation is presently empirical; Lean formalization would require routine `native_decide`-style enumeration. Data: `data/hard_residue_characterization.json`.

Proposition 5.8 (Branch-level hard set, Lean-verifiable). *Among the 150 nominal level-1 branch-states (r, c) with $r \in \text{hardResidues5}$ and $c \in \text{Fin } 30$, exactly 18 pairs genuinely resist closure at $\text{budget} \leq 3$:*

$$\begin{aligned} S_{18} = \{ & (0, 0), (0, 8), (0, 10), (0, 18), (0, 20), (0, 28), \\ & (1716, 6), (1716, 16), \\ & (4862, 12), (4862, 22), \\ & (16302, 0), (16302, 2), (16302, 10), (16302, 12), (16302, 20), (16302, 22), \\ & (17160, 0), (17160, 18) \}. \end{aligned}$$

The remaining 132 pairs admit a $\text{budget} \leq 3$ inadmissibility certificate covering some operative prime ($\text{budget} \leq 1$ directly, or covering of $\mathbb{Z}/p\mathbb{Z}$ at $p \in \{2, 3, 5\}$ by $\text{budget} \leq 2$ or $\text{budget} \leq 3$ forms). The cardinality claim $|S_{18}| = 18$ and the residue-distribution claim $6 + 2 + 2 + 6 + 2$ are Lean-verifiable (`hardBranches18_card`, `hardBranches18_distribution` in `Erdos647ReductionChain.lean`, both by `decide`).

Remark 5.9 (Distribution asymmetry). The 18 tentative hard branches distribute asymmetrically across the 5 hard residues: 6 each at $r = 0$ and $r = 16302$, and 2 each at $r = 1716$, $r = 4862$, $r = 17160$. The tentative-covering prime and required budget cap break down as: 11 branches at ($p = 2$, budget ≤ 6); 5 branches at ($p = 3$, budget ≤ 4); 2 branches at ($p = 2$, budget ≤ 4). Whether the $r = 0/r = 16302$ concentration reflects a deeper arithmetic feature distinguishing these two residues from the other three hard residues is an open question. Enumeration computed in `scripts/higher_budget_enumeration.py`; data recorded in `data/tentative_hard_branches.json`.

Remark 5.10 (Scope of pre-analytic methods). We tested three natural refinements that a sieve-theoretic attack might attempt against the 18 tentative branches, each designed to extract additional structure from the reduction without introducing new analytic hypotheses. Each is ruled out; together they characterize the boundary of pre-analytic methods on this set.

(i) *Finer branching at primes beyond mod30*. Moduli $M_{\text{new}} \in \{210, 510, 570, 714, 798, 1938\}$ (incorporating primes 7, 17, 19 in various combinations) were tested on each of the 18 tentative branches. Under no such M_{new} do all $M_{\text{new}}/\gcd(M_{\text{new}}, 30)$ sub-branches close rigorously at budget ≤ 3 . Richer branching produces partial closures — for example, $(r, c) = (17160, 0)$ under $M_{\text{new}} = 1938$ closes 18 of its 64 sub-branches rigorously, but the remaining 46 stay tentative. Consistent with the barrier theorem (Section 10): no finite prime-by-prime descent at primes greater than the clause count can eliminate a tentative branch.

(ii) *Shape restriction via square-elimination of budget-3 forms*. A budget-3 form $L(u) = \alpha u + \beta$ admits two arithmetic shapes: $L = p$ (prime) or $L = p^2$ (prime square). The p^2 option can be ruled out uniformly in u whenever there is a prime q with $q \mid \alpha$, $q \nmid \beta$, and $\beta \pmod q$ a non-zero quadratic non-residue modulo q ; then $L(u) \equiv \beta \pmod q$ is constant and a non-residue, hence never a square. For every budget-3 form in every one of the 18 tentative branches such a witness prime q exists (8/8 through 12/12 forms eliminated per branch, 100% elimination rate in all 18; see `scripts/qnr_elimination_prototype.py`). The effective form pool is thus reduced to budget ≤ 2 throughout; yet no branch closes, because the budget- ≤ 2 forms still fail to cover $\mathbb{Z}/p\mathbb{Z}$ at any operative prime in $\{2, 3, 5\}$. This rules out the entire family of square-based enhancements — QNR, Legendre symbol constraints, discriminant obstructions, any variant preventing L from being p^2 — because even the strongest possible outcome of such an analysis leaves the budget- ≤ 2 coverage deficit unchanged. The closure obstruction is localized to the budget-2 arithmetic itself.

(iii) *Sub-branching on the covering prime itself*. For each tentative branch with tentative covering prime $p \in \{2, 3\}$, sub-branching mod $30p$ (mod 60 for $p = 2$, mod 90 for $p = 3$) causes natural promotion of some forms to effective budget 1 under the reduction pipeline, yielding direct contradictions in certain sub-branches. At $p = 3$ exactly 1 of the 3 sub-branches closes this way per parent; at $p = 2$, 0 of the 2 sub-branches close. Since parent closure requires all sub-branches to close, no parent closes (`scripts/subbranch_promotion_prototype.py`, data in `data/subbranch_promotion_attempts.json`). This rules out the most natural "branch the deficit" attack as well.

(iv) *Residue-level closure rates under operative versus non-operative prime branching*. A finer-grained extension of (i). At $M_{\text{new}} = 30 \cdot 17 \cdot 19 = 9690$ (adding both operative primes dividing M), 488 of the 5,814 level-2 sub-branches close (8.39%), with a sharp residue-dependent hierarchy: $r = 0$ at 0.00%, $r = 16302$ at 5.88%, $r = 1716$ at 10.84%, $r = 4862$ at 18.73%, $r = 17160$ at 28.33%. The immunity of $r = 0$ has an arithmetic explanation: $2520r - k = -k$ forces $\gcd(2520M, |2520r - k|) = \gcd(2520, k)$ (every k in the 12-shape set is at most 24, hence coprime to $M = 11 \cdot 13 \cdot 17 \cdot 19$), so the base gcd contains no factor of M , $a_{\text{orig}} = 2520M/g$ absorbs all of M , and the extra-gcd peeling step has no M -prime factors to absorb under refinement at

primes dividing M . Replacing the operative pair $\{17, 19\}$ with non-operative $\{23, 29\}$ bypasses the obstruction: at $M_{\text{new}} = 30 \cdot 23 \cdot 29 = 20010$, 10,827 of 12,006 sub-branches close (90.18%), and the per-residue hierarchy collapses to a tight 87.96%–93.40% band across all five hard residues (in particular $r = 0$ rises from 0% to 87.96%). The finding generalizes to other non-operative pairs: at $M_{\text{new}} = 30 \cdot 31 \cdot 37 = 34,410$, 16,526 of 20,646 sub-branches close (80.04%) with the per-residue range 75.62%–83.70%. Smaller non-operative primes are more effective (each prime gives richer $\gcd(a_u, b_u)$ structure per unit j), but the mechanism is not specific to $\{23, 29\}$.

This shows that the residue-level variation observable at mod-9690 is a reflection of M -factor availability in the reduction pipeline, not an intrinsic hardness hierarchy among the five hard residues; the $(r \bmod 17, r \bmod 19)$ pair structure from Proposition 5.5 predicts the mod-9690 response but not the mod-20010 or mod-34410 response.

Residual structure at $M_{\text{new}}=20010$. The 1,179 open sub-branches left by non-operative branching have a sharp mod-23 signature: of the 23 residue classes $j \bmod 23$ (where j is the level-2 branching index), the class $j \equiv 8 \pmod{23}$ has 65.5% of its members universally-closing (closing at every one of the 18 tentative (r, c)), while $j \equiv 7 \pmod{23}$ has 0% and is the unique class with no universally-closing member. The mechanism: $30 \equiv 7 \pmod{23}$, so the 23-factor peeling in $\gcd(a_u, b_u) = \gcd(20010 a_{\text{orig}}, a_{\text{orig}}(30j + c) + b_{\text{orig}})$ fires iff j lies in a specific class mod 23 determined by (r, c, k) ; the class $j \equiv 7 \pmod{23}$ is precisely the one receiving no hit across the twelve shapes. The residual hardness of non-operative branching is thus a single CRT bottleneck, paralleling the $r = 0$ immunity at mod-9690 but at the level-2 index rather than the residue.

Consistent with (i), no single M_{new} closes every sub-branch of any tentative (r, c) , so no full tentative branch is eliminated by branching alone; (i)’s conclusion — that no finite prime-by-prime descent at primes greater than the clause count can eliminate a tentative branch — is reaffirmed. Experimental details: `scripts/a1_branch_323.py`, `scripts/a1b_branch_667.py`, `scripts/a1d_branch_1147.py`, `scripts/a1b_residual_structure.py`; `memos/docs/a1_branch323_result.md`, `docs/a1b_branch667_result.md`, `docs/c1_pair_partition.md`, `docs/a1b_residual_structure.md`.

Together, (i)–(iv) exhaust the natural pre-analytic routes to closing the 18 tentative branches. The closure obstruction localizes to the budget- ≤ 2 coverage at operative primes $\{2, 3, 5\}$, which no combination of finer branching (at operative or non-operative primes), form-shape restriction, or sub-branching on the covering prime can affect. Further progress on this set appears to require analytic input, specifically results of the form discussed in Section 11 — a density-level control on the budget- ≥ 4 Dickson-set tail, with a tuple-uniform Titchmarsh-type divisor bound as the nearest proxy target, or a prime-core-relative analog of Matthiesen’s affine-linear correlation estimates on the 13-form Dickson set.

6 The mod-11 square-kill and divisor-extension theorem

The hard-set characterization of Section 5 leaves a congruential residue of open content: for each hard $r \in \{0, 1716, 4862, 16302, 17160\}$ we know the 12-shape package is budget- ≤ 3 -admissible on the progression $N = Mt + r$, but the package is not pinned to primality until we close the mod-11 shape classification. This section supplies that pinning via a standalone arithmetic lemma on integers $x \equiv -1 \pmod{11}$, derives a divisor-extension corollary that applies to Erdős #647 candidates on hard residues, and proves a completeness theorem: under the budget bound $B_k \leq 5$, the resulting list of forced primality constraints consists of exactly one universal extension and six residue-specific

quotient-prime extensions.

6.1 The mod-11 arithmetic lemma

Throughout, write $M = 46189 = 11 \cdot 13 \cdot 17 \cdot 19$ and, for $k \mid 2520$, set $d_k := 2520/k$ and

$$x_k(N) := d_k \cdot N - 1, \quad B_k := \lfloor (k+2)/\tau(k) \rfloor.$$

Lemma 6.1 (Mod-11 rigidity). *Let $x > 1$ be an integer with $x \equiv -1 \pmod{11}$.*

1. *If $\tau(x) \leq 3$, then x is prime.*
2. *If $q \in \{17, 19\}$, $q \mid x$, and $\tau(x) \leq 5$, then $v_q(x) = 1$ and x/q is prime.*

Proof. (i) An integer with $\tau \leq 3$ has shape in $\{1, p, p^2\}$. The quadratic residues mod 11 are $\{1, 3, 4, 5, 9\}$; the class $-1 \equiv 10 \pmod{11}$ is a non-residue. Hence x is never a perfect square, ruling out p^2 , while $x > 1$ excludes 1. So $x = p$ is prime.

(ii) An integer with $\tau \leq 5$ divisible by $q \in \{17, 19\}$ has shape q^e for some $1 \leq e \leq 4$, or $q \cdot p$ with p prime and $p \neq q$. Direct computation gives

$$\begin{aligned} 17^e \pmod{11} &: 6, 3, 7, 9 & (e = 1, 2, 3, 4), \\ 19^e \pmod{11} &: 8, 9, 6, 4 & (e = 1, 2, 3, 4); \end{aligned}$$

none of these equals $10 \equiv -1$. The shapes q^e ($1 \leq e \leq 4$) are therefore excluded by $x \equiv -1 \pmod{11}$. The only remaining possibility is $x = q \cdot p$, whence $v_q(x) = 1$ and $x/q = p$ is prime. $\square$

6.2 Divisor-extension on Erdős candidates

Write $K_{\text{pure}} := \{1, 2, 3, 4, 6, 12\}$ for the divisors of 2520 with $v_p(k) < v_p(2520)$ for every prime $p \mid k$.

Corollary 6.2 (Divisor-extension at hard residues). *Fix a hard residue*

$$r \in \{0, 1716, 4862, 16302, 17160\},$$

and let $N = Mt + r$ for some $t \geq 0$. Suppose $n = 2520N$ is an Erdős #647 candidate.

1. *For $k \in K_{\text{pure}}$, $\gcd(k, x_k(N)) = 1$ unconditionally; hence $\tau(n - k) = \tau(k) \cdot \tau(x_k(N))$ and the Erdős inequality at k rewrites as $\tau(x_k(N)) \leq B_k$.*
2. *$x_k(N) \equiv -1 \pmod{11}$ for every $k \mid 2520$.*
3. *For $k \in K_{\text{pure}}$ with $B_k \leq 3$, $x_k(N)$ is prime. These are the six core budget- ≤ 3 forms with $d \in \{2520, 1260, 840, 630, 420, 210\}$.*
4. *For $k \notin K_{\text{pure}}$, coprimality can fail and the budget constraint combined with the per-prime case analysis of Section 3 gives the shape conditions recorded there.*
5. *For $k = 20$ ($d_{20} = 126 = 2 \cdot 3^2 \cdot 7$), coprimality $\gcd(20, x_{20}(N)) = 1$ holds except on the sub-progression $N \equiv 1 \pmod{5}$ (Case B), where $5 \mid x_{20}(N)$. Case A ($N \not\equiv 1 \pmod{5}$) forces $x_{20}(N)$ prime; Case B forces $x_{20}(N) = 5 \cdot q$ with q prime and $q \equiv 2 \pmod{11}$.*

Remark 6.3 (Erdős candidacy is a genuine hypothesis). The candidate assumption in Corollary 6.2 is necessary. At $N = M = 46189$ (a hard-residue survivor with $r = 0$ that is not an Erdős candidate) the form $x_2(N) = 1260 \cdot 46189 - 1 = 58,198,139 = 5297 \cdot 10987$ has $\tau(x_2) = 4 > 2 = B_2$. Survivorship alone gives the mod-11 congruence but not the τ -bound; candidacy supplies the bound.

6.3 Tightness of the budget bound

Proposition 6.4 (Tightness of $B_k \leq 5$). *The budget bound $B_k \leq 5$ in Corollary 6.2(v) and in Theorem 6.5 below is sharp: there exist hard-residue integers N and divisors $k \mid 2520$ with $B_k = 6$ for which $x_k(N)$ is divisible by $q \in \{17, 19\}$ yet $x_k(N)/q$ is composite, while the local Erdős inequality at that single offset k is still satisfied.*

Proof. Take $r = 1716$, $k = 72$, $d_k = 35$, $B_{72} = \lfloor 74/\tau(72) \rfloor = \lfloor 74/12 \rfloor = 6$, and $t = 42$. Then $N = Mt + r = 46189 \cdot 42 + 1716 = 1,941,654$, and $x_{72}(N) = 35N - 1 = 67,957,889 = 19^2 \cdot 188,249$ with 188,249 prime. Thus $\tau(x_{72}) = 6 = B_{72}$, $x_{72} \equiv 10 \pmod{11}$, and the Erdős inequality at $k = 72$ is satisfied ($\tau(n - 72) = \tau(72)\tau(x_{72}) = 12 \cdot 6 = 72 \leq 74 = k + 2$). Yet $v_{19}(x_{72}) = 2$, so $x_{72}/19 = 19 \cdot 188,249$ is composite, violating the conclusion of Corollary 6.2 if it were extended to $B_k \leq 6$.

The failure is structural. At $\tau(x) = 6$ two new shapes appear beyond those admitted at $\tau(x) \leq 5$: p^5 and p^2q . (The shape pqr has $\tau = 8$ and is not in play.) Both new shapes are compatible with $x \equiv -1 \pmod{11}$: direct computation yields $17^5 \equiv 19^5 \equiv -1 \pmod{11}$ (admitting p^5 for $p \in \{17, 19\}$), and $q \cdot s^2 \equiv -1 \pmod{11}$ has solutions in s for each $q \in \{17, 19\}$. So mod-11 no longer forces the prime-quotient conclusion at $B_k = 6$. $\square$

6.4 Completeness at $B_k \leq 5$

Theorem 6.5 (Completeness of the divisor-extension list). *Among the 48 divisors $d \mid 2520$:*

1. *Exactly thirteen divisors d satisfy $B_{2520/d} \leq 3$:*

$$d \in \{105, 140, 210, 252, 280, 315, 420, 504, 630, 840, 1260, 2520\} \cup \{126\}.$$

The first twelve are the 12-shape sieve set (Section 3); $d = 126$ (from $k = 20$, $B_{20} = 3$) is the universal extension.

2. *Exactly ten divisors d satisfy $B_{2520/d} \in \{4, 5\}$: $d \in \{42, 60, 63, 70, 84, 90, 120, 168, 180, 360\}$. Among all pairs (r, d) with r hard and d in this list, the congruence $d \cdot r \equiv 1 \pmod{q}$ (for $q \in \{17, 19\}$) holds in exactly the following six cases:*

r	d	k	forced by
1716	84	30	17
1716	168	15	19
4862	180	14	19
16302	84	30	17
17160	70	36	19
17160	90	28	17

For each such (r, d, q) and any hard-residue Erdős candidate N with $N \equiv r \pmod{M}$, the form $x_k(N) = d \cdot N - 1$ is divisible by q and the quotient $x_k(N)/q$ is prime (possibly modulo a Case-B-type sub-progression analog of Corollary 6.2(v) at a prime other than q).

3. *No divisor-extension constraint arises for any (r, d) outside the thirteen universal and six residue-specific cases above.*

Proof. Part (i) is arithmetic: the thirteen divisors listed are those with $B_{2520/d} \leq 3$. Part (ii) enumerates the divisors $d \mid 2520$ with $B_{2520/d} \in \{4, 5\}$ and tests $dr \equiv 1 \pmod{17}$ and $dr \equiv 1 \pmod{19}$ across the 50 candidate pairs $(r, d) \in \text{hardResidues5} \times \{42, 60, \dots, 360\}$; the six cases tabulated are the complete list. Part (iii) follows from the exhaustion of $B_k \leq 5$: at $B_k \geq 6$ the conclusion can fail (Proposition 6.4), and divisors with $B_{2520/d} \leq 5$ are covered by (i) and (ii). The computational content is enumerated in `scripts/verify_omitted_forms.py`. $\square$

Remark 6.6 (BH-friendly formulation). Expanding $N = Mt + r$ in each forced-prime form and clearing the denominator where applicable, the Case A pure-primality constraints of Theorem 6.5 can each be written as “ $L(t)$ prime” for an explicit integer-valued linear form $L \in \mathbb{Z}[t]$. On the Case A sub-progression (complementary to $N \equiv 1 \pmod{5}$ at $r = 0$, and analogously elsewhere), the Erdős #647 candidate condition reduces on the hard residues to simultaneous primality of an explicit tuple of linear forms in t : 13 forms at $r = 0$, 14 at $r = 4862$, 14 at $r = 16302$, 15 at $r = 1716$, and 15 at $r = 17160$. This is the Bateman–Horn-friendly formulation used in Section 12.

7 The offset atlas

The Case-A/Case-B structure of Corollary 6.2 and the residue-specific extensions of Theorem 6.5 become transparent under a single coordinatization of the affine forms $F_{r,k}(t) := 2520(Mt + r) - k$, which we call the *offset atlas*.

Definition 7.1 (Atlas objects). Write $A := 2520 \cdot M$, $C_{r,k} := 2520 \cdot r - k$, $P_0 := \{2, 3, 5, 7, 11, 13, 17, 19\}$, and $a_p := v_p(A)$, $c_p(r, k) := v_p(C_{r,k})$. A prime $p \in P_0$ is *active* at (r, k) iff $c_p \geq a_p$; otherwise *passive*. Write $S(r, k)$ for the set of active primes.

For each active prime $p \in S(r, k)$, Hensel lifting produces a unique residue class $t_{p,j} \pmod{p^j}$ at each level $j \geq 1$ on which $v_p(F_{r,k}(t)) \geq a_p + j$. A *branch* is an exact-valuation specification $\lambda = (\lambda_p)_{p \in S(r,k)}$: at $\lambda_p = j$, we require $t \equiv t_{p,j} \pmod{p^j}$ and $t \not\equiv t_{p,j+1} \pmod{p^{j+1}}$ (so $v_p(F_{r,k}(t)) = a_p + \lambda_p$ exactly). Combining across active primes via CRT pins t modulo $m_\lambda := \prod_{p \in S(r,k)} p^{\lambda_p}$. After reparameterizing $t = t_\lambda + m_\lambda u$, we obtain the *primitive cofactor* $Z_{r,k,\lambda}(u) = \alpha u + \beta \in \mathbb{Z}[u]$ with $\gcd(\alpha \cdot u + \beta, A) = 1$ for u in the branch.

Proposition 7.2 (Atlas factorization). *On branch λ of (r, k) ,*

$$F_{r,k}(t) = \left(\prod_{p \notin S} p^{c_p} \right) \cdot \left(\prod_{p \in S} p^{a_p + \lambda_p} \right) \cdot Z_{r,k,\lambda}(u),$$

and

$$\tau(F_{r,k,\lambda}(t)) = T_{r,k}^{\text{fix}} \cdot \left(\prod_{p \in S} (a_p + \lambda_p + 1) \right) \cdot \tau(Z_{r,k,\lambda}(u)),$$

where $T_{r,k}^{\text{fix}} := \prod_{p \notin S} (c_p + 1)$. Consequently the branch budget

$$B_{r,k,\lambda} := \left\lfloor (k + 2) / \left(T_{r,k}^{\text{fix}} \prod_{p \in S} (a_p + \lambda_p + 1) \right) \right\rfloor$$

bounds $\tau(Z_{r,k,\lambda}(u))$ whenever N is an Erdős #647 candidate.

The atlas is finite: restricting to (r, k, λ) with $B_{r,k,\lambda} \geq 2$, for r hard and $k \in [1, 100]$, produces 500 (r, k) entries with a total of 3,471 branches. At each branch the atlas also records the residues $Z_{r,k,\lambda}(u) \pmod{p}$ for $p \in P_0$ (fixed when $\alpha \equiv 0 \pmod{p}$, otherwise varying with u), and the list of exponent-patterns of $Z_{r,k,\lambda}(u)$ compatible with these residues via the order-subgroup realizability criterion in $(\mathbb{Z}/p)^\times$. The machine-readable dataset is in `data/offset_atlas.json`.

Remark 7.3 (Case A vs Case B at $k = 20$). At $(r, k) = (0, 20)$, Proposition 7.2 yields $S = \{5\}$, $T^{\text{fix}} = 3$, and two admissible branches with $\lambda_5 \in \{0, 1\}$. Case A ($\lambda_5 = 0$) has budget 3 and unique surviving pattern [1]; Case B ($\lambda_5 = 1$) has budget 2 and $Z \equiv 2 \pmod{11}$. This reproduces Corollary 6.2(v) directly from the atlas.

8 Empirical killer sets

The atlas classification (Section 7) together with the Erdős inequality at each $k \geq 1$ suggests an empirical investigation: for each hard residue r , is there a small set $S_r \subseteq [1, K_{\max}]$ such that every candidate N with $N \equiv r \pmod{M}$ fails the Erdős inequality at some $k \in S_r$?

Definition 8.1 (Base-enriched offsets). For residue r , the base-enriched k -set is the set of k -values whose affine form participates in the enriched tuple of Theorem 6.5; explicitly,

r	base-enriched k -set
0	$\{1, 2, 3, 4, 5, 6, 8, 9, 10, 12, 18, 20, 24\}$
1716	$\{1, 2, 3, 4, 5, 6, 8, 9, 10, 12, 15, 18, 20, 24, 30\}$
4862	$\{1, 2, 3, 4, 5, 6, 8, 9, 10, 12, 14, 18, 20, 24\}$
16302	$\{1, 2, 3, 4, 5, 6, 8, 9, 10, 12, 18, 20, 24, 30\}$
17160	$\{1, 2, 3, 4, 5, 6, 8, 9, 10, 12, 18, 20, 24, 28, 36\}$

The *killer universe* at residue r is $[1, 100]$ minus the base-enriched k -set.

Definition 8.2 (Strong near-miss). For residue r with enriched-tuple size m_r , a sample $N = Mt + r$ is a *strong near-miss* iff at least $m_r - 2$ of the m_r enriched-tuple forms are prime at N . The corresponding threshold values are 11, 13, 12, 12, 13 for $r \in \{0, 1716, 4862, 16302, 17160\}$ respectively.

Definition 8.3 (Killer set). For residue r , the *minimum killer set* S_r is the smallest subset of the killer universe at r such that for every observed strong near-miss N at r , some $k \in S_r$ satisfies $\tau(2520N - k) > k + 2$.

Proposition 8.4 (Small-cover phenomenon at $r = 0$). *Let $r = 0$ with the strong threshold of Definition 8.2 (primes ≥ 11 of 13 enriched forms). Freezing the empirical claims to the audited 301-record strong corpus used by `data/two_views_analysis.json` and `data/robustness_audits.json`, one has:*

1. *there exists a minimum killer set of size 3 covering all 301 records — specifically $\{15, 16, 27\}$ is the unique 3-element cover;*
2. *no 2-element cover exists;*
3. *the monitor set $S_{\text{mon}} = \{7, 11, 13, 16\}$ covers all 301/301 records. Across the combined strong-threshold corpus ($n = 303$ pooled $r = 0$, $r = 1716$, $r = 4862$), the monitor set still covers every record, and the $\{15, 16, 27\}$ cover remains universal on the $r = 0$ slice.*

Remark 8.5 (Exact minima are data-dependent; small-cover and monitor phenomena are stable). Three corpus-growth events demonstrate that specific minimum covers shift while the small-cover phenomenon persists. At an earlier snapshot with $n = 228$ strong near-misses, the unique size-3 cover was $\{14, 15, 16\}$. At the $n = 301$ snapshot, $\{14, 15, 16\}$ covered 300/301 (outlier at $t = 3,336,277,668,150$), so the minimum shifted to $\{15, 16, 27\}$. We freeze the paper’s empirical statements at this audited $n = 301$ / combined $n = 303$ snapshot to avoid mixing later telemetry into the theorem-facing text. The robust structural claim is not the identity of any specific minimum cover but the persistence of *some* small cover together with $\{7, 11, 13, 16\}$ monitor coverage on the frozen audited corpus. We do not treat either $\{14, 15, 16\}$ or $\{15, 16, 27\}$ as a theorem target; the theorem-shaped analog is a branch-dichotomy (see Remark 8.7).

Remark 8.6 (Cross-residue data: pilot only). At $r = 1716$ and $r = 4862$, the current strong-threshold corpus contains 1 record each (from Hetzner searches on $t \in [1, 10^9]$). At $r = 16302$ and $r = 17160$, the strong-threshold corpus is currently empty. We do not state per-residue minimum killer sets at these residues as formal propositions; the available data are pilot-level only. The monitor-coverage claim in Proposition 8.4(iii) includes these residues and remains 1/1 at each, consistent with (but not statistically equivalent to) the $r = 0$ result.

Remark 8.7 (Conditioning layer, diagnostic observation, and structural target). The outlier $t = 3,336,277,668,150$ is at the *strong near-miss* layer (11 of 13 enriched forms prime; enriched failures at $k = 18, 20$), not the *true base-pass* layer (13 of 13 enriched forms prime). The distinct conditioning layers form distinct stochastic regimes; the two should not be conflated.

At the strong near-miss layer, branch classification of the 301 records (`data/branch_classification.json`) shows exactly one record passes $\{14, 15, 16\}$ simultaneously; this record is killed by $\{7, 11, 13, 27\}$. Empirically at this layer,

$$\text{NearBasePass}_{11/13}(N) \wedge \bigwedge_{k \in \{14, 15, 16\}} \tau(2520N - k) \leq k + 2$$

has observed density $1/301 \approx 0.3\%$, and in the one observed instance failure occurs at smaller offsets rather than at $\{14, 15, 16\}$. This is a diagnostic observation at the strong near-miss layer, not a theorem-shaped claim about true candidates.

At the true base-pass layer (13/13), the observed hit at $t = 11,741,230,040,310$ is killed by $\{14, 15, 16\}$ at every offset. The outlier gives no direct evidence at this layer.

The natural structural target is therefore not a finite-cover statement but a *conditioned divisor-distribution model*: for each omitted offset k and each prime ℓ , the local probability

$$\rho_{\ell,k} := \Pr(\ell \mid Z_{0,k}(t) \mid \mathcal{B} \text{ locally passes mod } \ell),$$

where $\mathcal{B}$ denotes the enriched $r = 0$ base system and $Z_{0,k}$ is the primitive cofactor from the atlas of Section 7, together with the pairwise correlations $\text{Corr}(\omega(Z_{0,k}), \omega(Z_{0,j}))$. By an Euler-product over $\ell \leq Y$, this determines a conditioned Erdős–Kac-type distribution for $\omega(Z_{0,k}(t))$, whose marginal and joint tails govern the Erdős-inequality pass probabilities. We leave the computation and consequences of this model to future work; the outlier is one calibration point.

We do not promote $\{14, 15, 16\}$ as structurally sufficient. Its observed strength (99.67% empirical dominance at the strong near-miss layer) appears to come from high conditional failure probability of omitted offsets with small divisor budgets, not from forced incompatibility.

Remark 8.8 (Threshold normalization). The strong-threshold choice (miss-count ≤ 2 relative to the enriched tuple) is important: at looser thresholds the minimum killer set becomes polluted by offsets k that correspond to base-enriched forms, artifacts of the incomplete base constraint rather than genuine killers. In particular, at threshold $\geq 10/15$ at $r = 1716$, the minimum cover $\{1, 13, 42, 72\}$ includes $k = 1$; since $2520N - 1$ is already one of the 15 enriched forms at $r = 1716$, samples failing at $k = 1$ are precisely samples that are not candidates at all. Filtering the killer universe to exclude base-enriched k -values and restricting to strong near-misses yields the clean estimates of Proposition 8.4.

Remark 8.9 (Status: empirical, not theorem). Proposition 8.4 is an empirical statement about observed records. We tested three finite-incompatibility programs for a proof of “every candidate at $r = 0$ fails the Erdős inequality at some $k \in \{14, 15, 16\}$ ”: a local-covering solver at primes $p \leq 200$ not in P_0 ; an extension adding core-form primality constraints at $p \leq 1000$; a further extension adding the pairwise-gcd identities $\gcd(n - a, n - b) \mid |a - b|$ (for $a, b \in \{14, 15, 16\}$, these give $\gcd = 1$ or $\gcd \mid 2$). Across all 54 atlas branch triples at $r = 0$, none of the three programs produced a contradiction. This is a strong negative result on three specific finite-combinatorial vectors; it does not rule out other finite proofs, and it locates the natural next step for a proof in the analytic direction — a density-level control on the budget- ≥ 4 Dickson-set tail, with a tuple-restricted Titchmarsh-type bound as the nearest proxy target, which we discuss in Section 11. See `docs/theorem_sprint_conclusion.md` for the full record.

Remark 8.10 (Triage pipeline and trigger events). Proposition 8.4 is maintained as a live empirical claim by a four-stage triage pipeline applied to every new witness-search record: (i) base-enriched filter (drop any record failing the enriched tuple at its residue, i.e., below strong threshold); (ii) monitor filter against $S_{\text{mon}} = \{7, 11, 13, 16\}$; (iii) full killer-universe scan over $[1, 100]$ minus base-enriched offsets; (iv) deep-survivor classification. A *trigger event* is any record that passes stages (i) and (ii) — a strong near-miss that is not killed by the monitor. On the frozen audited combined strong corpus (total $n = 303$ records across $r = 0$, $r = 1716$, and $r = 4862$), the number of trigger events is zero: every strong-threshold record at every residue fails the Erdős inequality at some $k \in S_{\text{mon}}$. A single trigger event would require revisiting the paper’s empirical framing; continued null returns tighten it. Details in `scripts/analyze_two_views.py` and `docs/holding_pattern_rules.md`.

8.1 Branchwise effective budgets explain the small-offset obstruction profile

We now identify the structural mechanism governing the small-offset obstruction profile documented above. The claim is that the empirical pass/fail behavior at each offset k is controlled by an atlas-derived quantity, the *branchwise effective budget*, and that this quantity predicts the pass-rate ordering across the entire killer universe $k \leq 100$ at $r = 0$ and explains the cross-residue universality of the monitor set $\{7, 11, 13, 16\}$.

Definition 8.11 (Branchwise effective budget). On each atlas branch λ of (r, k) (Section 7), the atlas factorization

$$2520N - k = A_{r,k,\lambda}(t) \cdot Z_{r,k,\lambda}(t),$$

with $A_{r,k,\lambda}(t) = (\prod_{p \notin S} p^{c_p}) \cdot (\prod_{p \in S} p^{a_p + \lambda_p})$ the fixed-and-active divisor content and $Z_{r,k,\lambda}$ the primitive cofactor coprime to $A = 2520M$, defines the *branchwise effective budget*

$$B_{r,k,\lambda}^{\text{eff}} := \lfloor (k + 2) / \tau(A_{r,k,\lambda}) \rfloor.$$

Lemma 8.12 (Branch criterion for Erdős pass/fail). *For an Erdős #647 candidate $N = Mt + r$ lying on atlas branch λ at (r, k) , the Erdős inequality $\tau(2520N - k) \leq k + 2$ holds if and only if*

$$\tau(Z_{r,k,\lambda}(t)) \leq B_{r,k,\lambda}^{\text{eff}}.$$

In particular: $B^{\text{eff}} = 0$ forces automatic failure; $B^{\text{eff}} = 1$ forces $Z = 1$, impossible for sufficiently large N ; $B^{\text{eff}} = 2$ forces Z prime or $Z = 1$; $B^{\text{eff}} = 3$ forces Z prime, $Z = p^2$, or $Z = 1$; $B^{\text{eff}} \geq 4$ admits multiple squarefree and prime-power shapes.

Proof. Immediate from multiplicativity of τ applied to the atlas factorization in Definition 8.11 together with $\gcd(A_{r,k,\lambda}, Z_{r,k,\lambda}) = 1$. $\square$

The first-order conditioned Euler-product model (computational check in `scripts/conditioned_divisor_model.py`) shows that the primitive cofactors $Z_{r,k}$ are nearly exchangeable under local base conditioning at ordinary primes: one has $\mu_k(Y = 10^4) \approx 1.23$ and $\sigma_k \approx 1.10$ uniformly for the test offsets $k \in \{7, 11, 13, 14, 15, 16, 27\}$ at $r = 0$. Thus the observed asymmetry of the small-offset obstruction profile is not explained by first-order local divisibility. The asymmetry enters through the branchwise fixed content $A_{r,k,\lambda}$ in Definition 8.11.

Proposition 8.13 (Audited effective-budget distribution at $r = 0$). *For the 301-record strong-near-miss corpus at $r = 0$ (primes ≥ 11 of the 13 enriched forms), the branchwise effective-budget distribution at each test offset is:*

k	$B^{\text{eff}}=1$	2	3	4	5	6	7	<i>total</i>
7	3	7	32	259	0	0	0	301
11	0	0	0	23	0	278	0	301
13	0	0	0	0	16	0	285	301
14	1	53	0	247	0	0	0	301
15	5	161	0	135	0	0	0	301
16	7	59	234	0	0	0	0	300*
27	0	0	5	27	71	0	198	301

* one additional record at $k = 16$ has $B^{\text{eff}} = 0$ — fixed-content divisor count already exceeds $k + 2$, so Erdős inequality automatically fails on that branch.

In this corpus: every branch at $k = 16$ has $B^{\text{eff}} \leq 3$ (so Z must lie in $\{1, \text{prime}, p^2\}$); 55% of branches at $k = 15$ have $B^{\text{eff}} \leq 2$ (so Z must lie in $\{1, \text{prime}\}$); 18% of branches at $k = 14$ have $B^{\text{eff}} \leq 2$; and no branches at $k \in \{11, 13\}$ have $B^{\text{eff}} \leq 4$.

Proposition 8.14 (Severity ordering across $k \leq 100$ at $r = 0$). *Let p_k^{pred} denote the branch-averaged prediction*

$$p_k^{\text{pred}} = \sum_{\lambda} w_{k,\lambda} \cdot F(B_{r,k,\lambda}^{\text{eff}}),$$

where F is a Landau-order τ -pass approximation and $w_{k,\lambda}$ is a branch-weight measure. Let p_k^{emp} denote the empirical pass rate on the 301-record strong corpus at $r = 0$. Across all 87 offsets $k \in [1, 100]$ not in the $r = 0$ base-enriched set, the Spearman rank correlations are:

$$\rho(p^{\text{pred}}, p^{\text{emp}}) = 0.9386 \text{ (empirical branch weights), } 0.9330 \text{ (neutral-sampled branch weights).}$$

Empirical and neutral predictions themselves agree at $\rho = 0.9780$. Chronological splitting of the corpus into halves gives between-halves $\rho = 0.9639$, with neutral-prediction-vs-half correlations of 0.9385 and 0.9205 respectively. A bootstrap of 200 resamples for the neutral-prediction audit gives median $\rho = 0.9301$ with 95% confidence interval $[0.9218, 0.9375]$.

Remark 8.15 (Scope of validation). Proposition 8.14 validates the effective-budget mechanism at the $\mathcal{N}_{11/13}$ strong-near-miss layer across all offsets $k \leq 100$ in the frozen audited $r = 0$ corpus, under both empirical and neutral branch weights, and robustly to chronological partitioning and bootstrap. At the true base-pass layer $\mathcal{B}_{13/13}$ only the single confirmed witness at $t = 11,741,230,040,310$ exists, so this layer is not independently validated. The absolute magnitudes of the Landau prediction are approximately a factor of 1.5–2 below empirical rates, reflecting the crudeness of the first-order tail law; the ordering is stable. The main residuals occur at $k = 30, 36$, where a $B^{\text{eff}} = 4$ branch with surviving pattern $[1, 1]$ (semiprime) dominates the empirical distribution while the Landau approximation underweights the semiprime contribution. This indicates that effective budget is the first-order mechanism; pattern-level cofactor modeling is the natural second-order correction.

Proposition 8.16 (Cross-residue atlas tightness of $k = 16$). *Neutral-sampled effective-budget statistics across all five hard residues give the following values for $k = 16$:*

r	mean B^{eff}	fraction with $B^{\text{eff}} \leq 3$	rank among 87-86 non-base offsets
0	3.38	49%	1
1716	3.35	51%	3
4862	3.35	51%	2
16302	3.37	49%	2
17160	3.37	49%	3

At every hard residue, $k = 16$ lies among the three atlas-tightest offsets. The $k = 16$ atlas branches at each residue have no $B^{\text{eff}} > 4$ entries (the active prime is 2 with $a_2 = 3$, constraining the branch exponent contribution to a narrow range). The independence-product pass-rate scan of Proposition 12.2 independently identifies $k = 16$ as a top-3 bottleneck at every residue: empirical per- k pass rates at $k = 16$ are 0.336, 0.358, 0.351, 0.351, 0.348 for $r \in \{0, 1716, 4862, 16302, 17160\}$ respectively, confirming the atlas-geometric prediction at the observational level.

Remark 8.17 (Moving- K window empirical scan). Per- k empirical pass rates at $n = 1,000$ samples per (r, k) (Hetzner cpx51, 16-worker parallel run) reveal that the tight obstruction is concentrated at $k \leq 100$. Mean per- k pass rates at each hard residue, by K -block:

r	$[1, 100]$	$[101, 200]$	$[201, 500]$	$[501, 1000]$
0	0.806	0.958	0.992	0.999
1716	0.810	0.954	0.991	0.998
4862	0.805	0.959	0.991	0.999
16302	0.810	0.956	0.991	0.999
17160	0.808	0.954	0.991	0.998

Very-tight offsets (per- k pass rate < 0.3) appear only in $[1, 100]$ (0–2 per residue) and *never* in $[101, 1000]$. The deepest rate in $[101, 1000]$ is at $k = 120$ (≈ 0.50 at every residue). The fixed $k \leq 100$ screen captures the *sharp marginal* obstruction pressure; the moving tail at $k > 100$ does not introduce new individual lethal offsets.

Joint block survival. A complementary per- t experiment (Hetzner, $n = 500$ samples per residue, 16-worker parallel) tests the joint event $\bigcap_k \{\text{pass at } k\}$ within each K -block:

K -block	$r = 0$	$r = 1716$	$r = 4862$	$r = 16302$	$r = 17160$
[1, 100]	0.000	0.000	0.000	0.000	0.000
[101, 200]	0.006	0.004	0.008	0.004	0.012
[201, 500]	0.070	0.040	0.040	0.040	0.054
[501, 1000]	0.526	0.422	0.456	0.470	0.348

Marginal per- k rates near 1 compound to meaningful joint suppression: mean per- k rate 0.958 at [101, 200] gives naive product $0.958^{100} \approx 0.014$, matching observed 0.004–0.012; rate 0.999 at [501, 1000] gives $0.999^{500} \approx 0.61$, matching observed 0.35–0.53. *Cumulative* $P(\text{pass all } k \in [1, K])$ is 0/500 at every $K \in \{100, 200, 500, 1000\}$ across all residues.

Interpretation. The case-A screen at $k \leq 100$ captures the sharp marginal obstruction pressure; the moving tail at $k > 100$ contributes compound multiplicative suppression via many mild constraints, not via new individual lethal offsets. Both components are empirically non-trivial. See `scripts/moving_k_window_experiment.py`, `scripts/joint_block_survival_experiment.py`, `data/joint_block_survival_experiment.json`, and `docs/joint_block_survival_result.md`.

Remark 8.18 ($k = 14$ is a second stratum-invariant universal wall). An analogous stratum-invariance analysis to Remark 8.27 reveals that $k = 14$ shares the structural-wall property of $k = 7$:

stratum	sampled n	$k = 14$ pass rate	dominant B^{eff}
$C_{\geq 9}$	500	31.6%	$B = 4$ (85%)
$C_{\geq 10}$	500	39.4%	$B = 4$ (84%)
$C_{\geq 11}$	375	33.9%	$B = 4$ (83%)
$C_{\geq 12}$	23	21.7%*	$B = 4$ (70%)
$C_{\geq 13}$	1	0%*	$B = 2$
Monitor escapes	143	32.9%	$B = 4$ (87%)

Sieve-weighted ratio $R_{14}(y)$ for $y \in \{7, 11, 17, 29, 47, 97\}$ evaluates to

$$\{1.052, 1.052, 1.052, 0.994, 0.975, 0.983\},$$

essentially flat at 1. Base-tuple quality does not raise $k = 14$ survival. When $k = 14$ fails, dominant Z_{14} -shape is 3-distinct-primes ($\approx 45\%$ of fails), same as $k = 7$.

The joint structure is notable: at monitor-escape records, $k = 14$ pass rate is 32.9%, which equals the unconditional rate — confirming $k = 14$ is TRANSVERSE to monitor passage. Combined with the $k = 7$ result, we have TWO independent stratum-invariant universal walls. See `scripts/k14_transversality.py`.

Remark 8.19 (Monitor-escape family structure and $k = 14$ backup bottleneck). Of the 29,290 records in the $r = 0$ near-miss corpus ($C_{\geq 9}$), 143 pass the monitor $\{7, 11, 13, 16\}$. Their first-fail distribution concentrates heavily at $k = 14$:

first-fail k	count (of 143)
14	96 (67.1%)
15	38 (26.6%)
21	4
17	3
19, 22	1 each

At first-fail, the B^{eff} distribution is dominated by $B = 4$ ($97/143 = 67.8\%$) and $B = 2$ ($36/143 = 25.2\%$). At $B = 2$ the branch forces Z prime and the record fails because Z is semiprime or larger; at $B = 4$ the record's Z is a 3- or 4-prime integer exceeding budget $\tau \leq 4$. The dominant Z -shape at first-fail is 3-distinct-primes ($65/143 = 45.5\%$).

Secondary monitor coverage. Across the 143 monitor-escape records, empirical catch rates are:

- $\{14, 15, 27\}$: $138/143 = 96.5\%$.
- $\{15, 16, 27\}$: $129/143 = 90.2\%$.
- $\{14, 16\}$: $96/143 = 67.1\%$.
- $\{27\}$ alone: $87/143 = 60.8\%$.

The combined network $\{7, 11, 13, 14, 15, 16, 27\}$ catches $138/143$ monitor-escapes at low k ; the remaining 5 pass the entire network and first-fail at slightly higher k (primarily $k = 17$, $k = 21$, or at $k \in [101, 1000]$). The emergent structural statement is therefore: *the robust monitor is a low-offset network, not any single minimum cover*. Specific 3-element covers continue to drift with corpus growth; the network $\{7, 11, 13, 14, 15, 16, 27\}$ has been stable across the $n = 301, 330, 375$ snapshots. See `scripts/monitor_escape_family_classification.py`, `scripts/batch_autopsy_143_escapes.py` and `data/batch_autopsy_143_escapes.json`.

Remark 8.20 (Moving-tail is MORE suppressive conditional on monitor-escape). The unconditional moving-tail block survival of Remark 8.17 measures $\Pr(\text{pass block} \mid \text{raw } t)$. The sharper conditional question is:

$$\Pr(\text{pass block } [K_{\text{lo}}, K_{\text{hi}}] \mid \text{passes monitor } \{7, 11, 13, 16\}).$$

Using the 143 observed monitor-escape records as a conditional sample:

K -block	raw Pr	S_1 (monitor-escape) Pr	ratio S_1/raw
[101, 200]	0.006	0.000	0.00
[201, 500]	0.070	0.014	0.20
[501, 1000]	0.526	0.294	0.56

Conditioning on monitor-escape makes the moving-tail block survival *lower*, not higher: by factor 0.00–0.56 across K -blocks. Monitor-escape records are not tail-friendly; if anything, they carry additional modular structure that suppresses survival at $k > 100$ as well.

Further, among the 143 monitor-escape records, only 5 pass the full network

$$\{7, 11, 13, 14, 15, 16, 27\}.$$

All 5 first-fail at $k \in \{17, 21\}$, all within $[1, 100]$; the full $k \leq 100$ screen therefore catches all 143 monitor-escapes (no record in the current corpus requires the moving tail for catching). The moving-tail analysis above provides additional multiplicative suppression beyond the screen, not the first catch. See `scripts/conditional_moving_tail_on_escapes.py` and `scripts/deep_escape_ledger.py`.

Remark 8.21 (Deep-network escape ledger and $k = 21$ as a third wall). The 5 records passing the full network $\{7, 11, 13, 14, 15, 16, 27\}$ are:

t	primes	first-fail k	τ	bound
256,333,196,208	9	21	48	23
314,789,936,208	9	17	192	19
508,200,324,660	9	17	64	19
2,441,574,485,160	9	21	32	23
56,750,755,674,480	9	21	32	23

The deep escapes concentrate at $k = 21$ (3/5) and $k = 17$ (2/5). Analogous transversality analysis to Remarks 8.27 and 8.18 gives:

k	$C_{\geq 9}$	$C_{\geq 10}$	$C_{\geq 11}$	Monitor	Network	note
7	38.8%	36.8%	36.7%	36.5%	40%	wall, $B = 4$
14	31.6%	39.4%	33.9%	32.9%	60%	wall, $B = 4$
21	36.4%	36.6%	33.6%	30.8%	20%	wall, $B = 5$
17	69.8%	68.8%	71.2%	71.3%	60%	not a wall, $B = 9$

Three stratum-invariant walls: $k = 7$, $k = 14$, $k = 21$, each with pass rate ≈ 33 – 37% and B^{eff} dominated by $B = 4$ or $B = 5$. $k = 17$ is loose ($\sim 70\%$ pass rate, $B^{\text{eff}} = 9$ dominant) and serves as an occasional escape-catcher via rare $B = 6$ branches rather than as a wall.

The extended low-offset network $\{7, 11, 13, 14, 15, 16, 17, 21, 27\}$ catches all 143 monitor-escapes and all 5 deep-network escapes in the current corpus. See `scripts/deep_escape_ledger.py` and `scripts/k17_k21_transversality.py`.

Lemma 8.22 (Shape decomposition). *Let $n \geq 1$. If $\tau(n) \leq 4$, then*

$$n \in \{1, p, p^2, p^3, pq\},$$

with p, q prime and, in the last case, $p < q$. Conversely, each of these forms has $\tau(n) \leq 4$.

Proof. Write $n = \prod_{i=1}^r p_i^{e_i}$. Then $\tau(n) = \prod (e_i + 1) \leq 4$ forces either $r = 0$ ($n = 1$); or $r = 1$ with $e_1 \in \{1, 2, 3\}$ ($n \in \{p, p^2, p^3\}$); or $r = 2$ with $e_1 = e_2 = 1$ ($n = pq$, $p \neq q$). $\square$

Theorem 8.23 (L2A wall asymptotic thinning). *Fix an affine form $L(u) = \alpha u + \beta$ with $\alpha \geq 1$, $(\alpha, \beta) = 1$, and prime support of α contained in $\{2, 3, 5, 11, 13, 17, 19\}$. In particular the statement applies uniformly to the three cases $(\alpha, \beta) \in \{(8, 314, 020, -1), (16, 628, 040, -1), (5, 542, 680, -1)\}$, which are the dominant-branch primitive cofactors at $k \in \{14, 7, 21\}$ respectively at the residue $r = 0$, and to their residue-translated analogues at the other four hard residues. Let*

$$S(U) := \#\{u \in [1, U] : \tau(\alpha u + \beta) \leq 4\}.$$

Then

$$S(U) \ll_{\alpha, \beta} U \cdot \frac{\log \log U}{\log U} \quad (U \rightarrow \infty),$$

and in particular $S(U) = o(U)$. There exists an effective $U_0 = U_0(\alpha, \beta)$ such that $S(U) \leq U/2$ for every $U \geq U_0$. Explicitly, one may take $U_0 = \lceil e^{300} \rceil < 2 \cdot 10^{130}$ uniformly for the three listed α .

Proof. Write $X := \alpha U + |\beta|$. By Lemma 8.22 every u contributing to $S(U)$ satisfies $\alpha u + \beta \in \{1, p, p^2, p^3, pq\}$. Accordingly

$$S(U) \leq 1 + N_p(U) + N_{p^2}(U) + N_{p^3}(U) + N_{pq}(U),$$

where $N_\bullet(U)$ counts $u \leq U$ such that $\alpha u + \beta$ takes the corresponding shape.

Perfect-power branches. If $\alpha u + \beta = p^2$ or p^3 , then $\alpha u + \beta \leq X$ forces $p \leq X^{1/2}$ or $p \leq X^{1/3}$ respectively, and each such p determines at most one u . Hence

$$N_{p^2}(U) \leq X^{1/2} \ll_{\alpha, \beta} U^{1/2}, \quad N_{p^3}(U) \leq X^{1/3} \ll_{\alpha, \beta} U^{1/3},$$

both $o(U)$.

Prime branch. The condition $\alpha u + \beta = p$ is equivalent to $p \in [2, \alpha U + \beta]$ and $p \equiv \beta \pmod{\alpha}$. Since $(\alpha, \beta) = 1$ this is a reduced residue class. By PNT in AP for fixed modulus α ,

$$N_p(U) = \frac{\text{Li}(\alpha U)}{\varphi(\alpha)} + o_\alpha\left(\frac{U}{\log U}\right) \sim c_\alpha \frac{U}{\log U}, \quad c_\alpha = \frac{\alpha}{\varphi(\alpha)}.$$

Semiprime branch. If $\alpha u + \beta = pq$ with $p < q$, then $p \leq \sqrt{X}$, and $p \nmid \alpha$ (since $p \mid \alpha$ combined with $p \mid pq = \alpha u + \beta$ would give $p \mid \beta$, contradicting $(\alpha, \beta) = 1$). For each such p the remaining prime q lies in the residue class $\beta p^{-1} \pmod{\alpha}$. Hence

$$N_{pq}(U) \leq \sum_{\substack{p \leq \sqrt{X} \\ p \nmid \alpha}} \pi\left(\frac{X}{p}; \alpha, \beta p^{-1}\right).$$

By PNT in AP uniformly over the finitely many reduced residue classes mod α ,

$$\pi\left(\frac{X}{p}; \alpha, a\right) = \frac{\text{Li}(X/p)}{\varphi(\alpha)} + o_\alpha\left(\frac{X/p}{\log(X/p)}\right).$$

For $p \leq \sqrt{X}$ one has $\log(X/p) \geq \frac{1}{2} \log X$, so $\text{Li}(X/p) \ll X/(p \log X)$. Inserting and using Mertens

$$\sum_{\substack{p \leq \sqrt{X} \\ p \nmid \alpha}} \frac{1}{p} = \log \log X + B_\alpha + o(1)$$

(with $B_\alpha = B_1 - \sum_{p \mid \alpha} 1/p$) yields

$$N_{pq}(U) \ll_\alpha \frac{X}{\varphi(\alpha)} \cdot \frac{\log \log X}{\log X} \asymp c_\alpha U \frac{\log \log U}{\log U}.$$

Summation. Combining,

$$S(U) \leq 1 + N_p(U) + N_{p^2}(U) + N_{p^3}(U) + N_{pq}(U) \ll_{\alpha, \beta} U \frac{\log \log U}{\log U} = o(U).$$

Effective U_0 . Replacing PNT in AP by the effective Montgomery–Vaughan Brun–Titchmarsh inequality $\pi(y; \alpha, a) \leq 2y/(\varphi(\alpha) \log(y/\alpha))$, which holds for every $y > \alpha$ with no exceptional-zero dependence, and retracing Steps 1–3 at $\log U \geq 300$, one verifies that $S(U)/U < 0.463$ for all $U \geq e^{300}$

uniformly in the three listed α . Thus $U_0 = \lceil e^{300} \rceil < 2 \cdot 10^{130}$ suffices. The constant is deliberately not optimised; sharper explicit Brun–Titchmarsh or PNT-in-AP inputs (Bennett–Martin–O’Byrne–Rechnitzer for $q \leq 10^5$, Ramaré for wider ranges at correspondingly larger x_0) lower U_0 , but the resulting threshold remains well beyond corpus scale. This is the analytic content of *Level 2A*; the corpus-scale effective version (*Level 2B*) is discussed in Remark 8.26. $\square$

Theorem 8.24 (L2B partial credit). *Under the hypotheses of Theorem 8.23, the sharper envelope*

$$S(U) \leq c_\alpha U \frac{\log \log(\alpha U) + 3}{\log(\alpha U)}$$

holds for all $U \geq 2 \cdot 10^{20}$. Consequently

$$S(U) \leq \frac{1}{2}U \quad \text{for every } U \geq U_0^* \text{ with } U_0^* = 6.6446042099 \cdot 10^{24}.$$

Proof sketch. Replace classical Brun–Titchmarsh with the Yamada (2023) explicit interval form

$$\pi(x+y; k, \ell) - \pi(x; k, \ell) < \frac{2y}{\varphi(k)(\log(y/k) + 0.8601)}, \quad y > k,$$

and retrace Steps 2–3 of the proof of Theorem 8.23 with the finite-range Mertens-in-AP bound

$$\left| \sum_{\substack{p \leq y \\ p \nmid \alpha}} \frac{1}{p} - \log \log y - B_\alpha \right| \leq \varepsilon, \quad \varepsilon = 3.88 \cdot 10^{-4} \ (y \geq 10^5),$$

derivable by direct sieving up to $y = 10^8$ and Dusart’s $|r(y)| \leq 0.2/\log^3 y$ for $y \geq 2,278,383$. The prime branch contributes

$$K_{\text{prime}}(U) = \frac{2 \log(\alpha U)}{\log U + 0.8601},$$

and the semiprime branch contributes

$$K_{\text{semi}}(U) \leq \log(\alpha U) \cdot T_{0.8601}(U) - \log \log(\alpha U),$$

with $T_\delta(U)$ the Yamada-kernel hyperbola sum. Direct computation at $U = 2 \cdot 10^{20}$ gives $K_{\text{prime}} = 2.6333$, $K_{\text{semi}} \leq 0.3434$, $K_{\text{cube}} < 10^{-10}$, totalling $K \leq 2.9767 < 3$. Both branches are monotonically decreasing for $U > 2 \cdot 10^{20}$. The threshold U_0^* is the numerical root of $c_\alpha(\log \log(\alpha U) + 3)/\log(\alpha U) = 1/2$. Full numerical details are in `docs/level2b_partial_credit_codex.md`. $\square$

Remark 8.25 (Why the corpus-scale $U_0 \leq 10^{13}$ is still open). At $U = 10^{13}$, the Yamada sharpening gives $K_{\text{prime}} = 2.979$ and $K_{\text{semi}} = 1.149$, total $K = 4.128$, well above the threshold $K \leq 0.75$ required for $S(U) \leq U/2$. Maynard’s η -gain for the Brun–Titchmarsh constant is triggered only when $\log x/\log q \geq 8$, i.e. $x \geq \alpha^8$; for the prime branch this requires $U \geq \alpha^7 \approx 2.75 \cdot 10^{48}$, and for the worst semiprime inner range $U \geq \alpha^{15} \approx 10^{104}$. Both are out of reach at corpus scale. Closing the corpus-scale gap thus requires a technique beyond the Maynard/Yamada Brun–Titchmarsh family — see Remark 8.26 for the surveyed alternatives.

Remark 8.26 (Analytic wall program: shape decomposition and rigour levels). The three stratum-invariant walls $W = \{7, 14, 21\}$ admit a clean analytic reduction. On the dominant branch $\lambda = (7 : 0)$, the primitive cofactor satisfies $Z_{r,k,(7:0)}(u) = \alpha u + \beta$ with $\alpha = A/k$, $\gcd(\alpha, \beta) = 1$. The corresponding B^{eff} is 4 for $k \in \{7, 14\}$ and 5 for $k = 21$. Passing the wall requires $\tau(Z) \leq B^{\text{eff}}$. Since Proposition 8.36 extends to the dominant $B^{\text{eff}} = 4$ branch by the same residue obstruction on β modulo the small primes, the shape $Z = p^2$ is excluded, and the target reduces to the *shape-decomposition lemma*:

$$\tau(Z) \leq 4, Z \neq p^2 \implies Z \in \{1, p, p^3, pq\},$$

which yields the explicit decomposition

$$S_k(U) \leq \pi(\alpha U + |\beta|; \alpha, \beta) + (\alpha U)^{1/3} + \sum_{\substack{p \text{ prime} \\ p \leq \sqrt{\alpha U} \\ \gcd(p, \alpha) = 1}} \pi\left(\frac{\alpha U + |\beta|}{p}; \alpha, \beta p^{-1}\right).$$

The remaining analytic question is the *rigorous constant* controlling the semiprime sum. Numerical experiments (`scripts/wall_decomposition_k14.py`, `scripts/wall_empirical_sample.py` and `scripts/mertens_correction_test.py`) at $\alpha = 8,314,020$ give:

- Prime-density heuristic matches empirical to $< 3\%$; the Euler-product factor $c_\alpha = \alpha/\varphi(\alpha) \approx 5.01$ is correct.
- The empirical semiprime constant at corpus scale fits at $c_{\text{fit}}(U) \approx 3.32$, i.e. $c_{\text{fit}}/c_\alpha \approx 0.66$. This *is not* an asymptotic Selberg–Delange leading constant; it is the finite-range effect of excluding the small primes dividing α from the semiprime harmonic sum. The cleanest statement uses the α -restricted prime-harmonic function

$$H_\alpha(y) := \sum_{\substack{p \leq y \\ p \nmid \alpha}} \frac{1}{p}.$$

Within a standard Mertens expansion $\sum_{p \leq y} 1/p = \log \log y + M_{\text{Mertens}} + o(1)$, the subtraction $\sum_{p|\alpha} 1/p = M_\alpha$ is dominant at our scales, giving the approximation

$$\frac{c_{\text{eff}}(U)}{c_\alpha} \approx 1 - \frac{M_\alpha}{\log \log(\alpha U)}, \quad M_\alpha = \sum_{p|\alpha} \frac{1}{p} = \frac{1}{2} + \frac{1}{3} + \frac{1}{5} + \frac{1}{11} + \frac{1}{13} + \frac{1}{17} + \frac{1}{19} \approx 1.313.$$

The simple first-order prediction $1 - M_\alpha/\log \log(\alpha U)$ sits roughly 6–9% *below* the empirically-measured ratio across $U \in [10^8, 10^{17}]$ (measured on a full $k \times r \times \log_{10} U$ matrix at 10^5 samples per cell using a Pollard-Rho shape classifier). The directional agreement — identical across the three walls $k \in \{7, 14, 21\}$ and matching the Mertens-correction prediction that all three walls share a structural constant — is the important content. The residual 6–9% gap is expected: the simple formula captures the dominant α -dependent subtraction M_α but omits lower-order Mertens/AP/boundary corrections. The exact analytic object is the finite prime-harmonic sum $H_\alpha(\sqrt{\alpha U})$ inside a one-sided semiprime envelope; the first-order expression is a leading approximation.

- Classical Brun–Titchmarsh with $c_{\text{mult}} = 2$ gives $U_0 \sim 10^{54}$ — unusable at corpus scale.

- Under the empirical finite-range Mertens-corrected envelope, the decomposition upper bound drops below $U/2$ (both pointwise and cumulatively) at $U_0 \approx 10^8$, well within the corpus scale. The analytic target consists of an *explicit one-sided upper envelope* for the semiprime sum of the form

$$N_{pq}(U; \alpha, \beta) \leq \frac{\alpha U}{\varphi(\alpha)} \sum_{\substack{p \leq \sqrt{\alpha U} \\ p \nmid \alpha}} \frac{1}{p \log(\alpha U/p)} \cdot (1 + \text{explicit error})$$

together with Brun–Titchmarsh-level bounds for N_p and the trivial bound for N_{p^3} . We do not claim this as a theorem; we state the target envelope and record that the empirical data support it at $U_0 \approx 10^8$ for all three walls.

We structure the rigour roadmap into five sub-levels:

Level 1 Empirical wall (achieved): pass rate $\approx 1/3$ at each $k \in W$, stratum-invariant to ± 4 pp.

Level 2A Unconditioned asymptotic thinning: $\#\{u \leq U : \tau(\alpha u + \beta) \leq 4\} = o(U)$. Reachable by fixed-modulus prime-number-theorem-in-AP plus the shape decomposition. Existence of U_0 follows but the explicit value is not controlled.

Level 2B Effective corpus-scale thinning: $\#\{u \leq U : \tau(\alpha u + \beta) \leq 4\} \leq U/2$ for all $U \geq U_0$ with explicit $U_0 \leq 10^{13}$. Blocked by the Brun–Titchmarsh factor-2 in classical upper-bound sieves. Using the character decomposition $\pi(y; \alpha, a) = \varphi(\alpha)^{-1} \sum_{\chi} \overline{\chi(a)} \pi_{\chi}(y)$, the principal contribution at $X = \alpha U \approx 8 \cdot 10^{19}$ ($U = 10^{13}$) is $P_2^{(0)}(X) \approx 3.13 \cdot 10^{12}$, already within the $c_{bt} \leq 1.25$ budget ($D_{\text{semi}} \approx 2.75 \cdot 10^{12}$, giving principal-only $c_{\text{eff}}^{(0)} = 1.140$). The theorem therefore reduces to bounding the nonprincipal contribution $|E_{\alpha}(X)|$ by $\approx 3.03 \cdot 10^{11}$. We surveyed the standard candidate methods: large-sieve aggregate ($\approx 1.86 \cdot 10^{16}$; misses by $\sim 6 \cdot 10^4$), Pólya–Vinogradov / Burgess on primes ($\approx 3.83 \cdot 10^{13}$; misses by ~ 126), dispersion method (no explicit finite-scale bound available), unconditional zero-density estimates (unable to rule out the exceptional-zero contribution at the required strength), and sharper Brun–Titchmarsh of Maynard and Yamada type (gives $\approx 8.8 \cdot 10^{12}$; misses by ~ 29). None supplies the required one-sided AP-semiprime envelope with constant $\lesssim 1.25$ at this X . We therefore do not pursue Level 2B unconditionally in this paper. A speculative bespoke approach — an explicit weighted $\Lambda * \Lambda$ character expansion, or certified segmented sieving over the $\varphi(\alpha) = 1,658,880$ invertible residue classes mod α , using the fact that $\alpha = 2^2 \cdot 3^2 \cdot 5 \cdot 11 \cdot 13 \cdot 17 \cdot 19$ has only 12 primitive-conductor divisors exceeding Platt’s $q \leq 400,000$ verified range — might yield the required finite constant as a computational project; we flag it as future work.

Level 3 Conditional wall lemma under the *base-conditioned local equidistribution* hypothesis $\text{BCLE}_r(Y)$: level-of-distribution $Y = U^{\theta}$ for some $\theta > 0$ on the base-tuple-conditioned set $\mathcal{B}_r(U)$ (no known plug-in theorem; cf. Maynard’s dense-cluster hypothesis and Benatar’s subset-BV framework; analogous in role to Elliott–Halberstam).

Level 4 Multi-wall cascade: combine Level-3 bounds at all three $k \in W$ with a controlled cross-offset correlation hypothesis; this is the level at which the escape-ladder heuristic approaches a conditional finiteness statement.

None of Levels 2–4 is claimed in this paper; the remark records the target structure and the empirical constants that make it plausible. See `docs/analytic_wall_k14.md`, `docs/bcle_`

hypothesis_formalization.md and docs/heuristic_finiteness_sketch_v2.md for the detailed programme.

Remark 8.27 ($k = 7$ is a stable bottleneck under conditioning). A complementary empirical finding at the $r = 0$ near-miss corpus: the per- k pass rate at $k = 7$ is approximately constant across conditioning strata $C_{\geq 9}, C_{\geq 10}, C_{\geq 11}$ (primes $\geq 9, 10, 11$ of the 13 enriched forms respectively). Sample rates:

stratum	sampled n	$k = 7$ pass rate	dominant B^{eff} at $k = 7$
$C_{\geq 9}$	500	38.8%	$B^{\text{eff}} = 4$ (87%)
$C_{\geq 10}$	500	36.8%	$B^{\text{eff}} = 4$ (83%)
$C_{\geq 11}$	330	36.7%	$B^{\text{eff}} = 4$ (86%)
$C_{\geq 12}$	21	57.1%*	$B^{\text{eff}} = 4$ (90%)
$C_{\geq 13}$	1	0%*	$B^{\text{eff}} = 4$

*Small-sample rates (binomial-compatible with the uniform $\sim 37\%$ seen in large strata; not separately informative at $n = 21$ and $n = 1$). The B^{eff} -branch distribution at $k = 7$ is also stable: dominated by $B^{\text{eff}} = 4$ with $\approx 10\%$ at $B^{\text{eff}} = 3$ and $\approx 2\%$ at $B^{\text{eff}} = 2$. Mean $\tau(2520N - 7)$ is ≈ 20 across strata. When $k = 7$ fails, the dominant $Z_7(t)$ -shape is a 3-distinct-prime integer ($\approx 50\%$ of fail cases), followed by 4-prime and 5-prime shapes.

This empirical stability of $k = 7$ across conditioning explains why the single confirmed 13-tuple witness at $t = 11,741,230,040,310$ fails at $k = 7$: the witness is $C_{\geq 13}$, but the $k = 7$ obstruction does not weaken with stronger base-tuple primality.

A complementary sieve-weighted surrogate confirms the transversality: define the base-quality score $Q_y(t) = \#\{d \in \mathcal{B} : dN - 1 \text{ has no prime factor } \leq y\}$, where $\mathcal{B}$ is the 13-form enriched tuple at $r = 0$; sample $n = 2,000$ at $t \in [10^8, 10^9]$ and compute the ratio $R_7(y) = \Pr(k = 7 \text{ pass} \mid Q_y \geq Q_y^{(75\%)}) / \Pr(k = 7 \text{ pass} \mid \text{raw})$. Staged across $y \in \{7, 11, 17, 29, 47, 97\}$, $R_7(y)$ takes values $\{0.971, 0.971, 0.971, 0.942, 0.936, 0.922\}$, all in $[0.92, 0.97]$. High base-quality samples have, if anything, a slightly *lower* $k = 7$ pass rate than raw; there is no evidence that base-tuple conditioning raises $k = 7$ survival. This is a positive falsification of the hypothesis that caveat (b) could close the remaining gap in the density model at $k = 7$. See `scripts/k7_sieve_weighted_surrogate.py`, `data/k7_sieve_weighted_surrogate.json`, `scripts/monitor_escape_k7_autopsy.py`, `data/monitor_escape_k7_autopsy.json`.

Remark 8.28 ($k = 16$ at the strongest-near-miss subset). At the $r = 0$ strong threshold with 12 or more primes of the 13 enriched forms ($n = 21$ records in the current corpus), $k = 16$ alone fails the Erdős inequality in every case: $\tau(2520N - 16) > 18$ for all 21 records at this threshold. At the broader $r = 0$ strong-near-miss telemetry snapshot (primes ≥ 11 , $n = 375$), $k = 16$ alone kills $\approx 87\%$ of records, with the monitor $\{7, 11, 13, 16\}$ required to cover the remaining 44. The 21/21 rate at the $k \geq 12$ subset is consistent with — though a stronger observation than — the atlas-tightness of Proposition 8.16. Sample size at the $k \geq 12$ layer is small; we report this as an empirical observation from a later telemetry snapshot, not as a universal claim.

Remark 8.29 (Structural explanation of the monitor $\{7, 11, 13, 16\}$). The universality of the monitor $\{7, 11, 13, 16\}$ decomposes into residue-generic and residue-specific components:

1. *Residue-generic*: $k = 16$ is atlas-tight at every residue (Proposition 8.16); $k = 7$ ranks within the top 8–10 tightest offsets at every residue (mean $B^{\text{eff}} \approx 3.83$ across all r).

2. *Residue-specific extreme-tightness*: $k = 13$ becomes extreme-tight at $r = 1716$ and $r = 16302$ (mean $B^{\text{eff}} \approx 2.85$, 100% of branches with $B^{\text{eff}} \leq 3$, rank 1–2), reflecting the active-prime 13 content at those residues; $k = 11$ becomes extreme-tight at $r = 17160$ (rank 2, 100% with $B^{\text{eff}} \leq 3$), reflecting the analogous active-prime 11 content.

Thus at every hard residue, at least one member of $\{7, 11, 13, 16\}$ lies in the top three atlas-tightest offsets. The monitor’s empirical coverage is consistent with this branch-budget geometry, though the formal coverage claim remains empirical.

Remark 8.30 (Alternate universal-tight sets and empirical universality). Audit 4 identifies $\{7, 16, 40, 72\}$ as the universal atlas-tight-top-10 intersection across all five hard residues — a candidate alternate monitor family. However, on the expanded strong-near-miss corpus of 330 records at $r = 0$ (primes ≥ 11 , updated 2026-04-18), $\{7, 16, 40, 72\}$ misses one record at $t = 2,445,307,576,020$ (99.70% empirical coverage), while the empirically-motivated monitor $\{7, 11, 13, 16\}$ still achieves 100%. A second three-element universal monitor, $\{15, 16, 27\}$, also achieves 100% coverage on the expanded corpus and is a candidate simplification; $\{7, 11, 13, 16\}$ remains the base monitor used throughout the paper because $k = 16$ alone covers 86.67% and each of $\{7, 11, 13\}$ contributes residue-specific lethality (see Remark 8.29). We flag these alternatives for subsequent analysis; none replaces the empirically-validated monitor here.

Remark 8.31 (Interpretation and scope). The branchwise severity ordering of omitted offsets is a theorem-shaped consequence of atlas arithmetic (Lemma 8.12); the empirical pass-rate distribution at each offset reflects this ordering under a Landau tail law for primitive cofactors. The empirical killer-set phenomenon and the cross-residue monitor universality follow from the branch-budget geometry. We do not claim that the monitor $\{7, 11, 13, 16\}$ provably covers every Erdős candidate, nor that the effective-budget mechanism constitutes a proof of finiteness for #647. What we claim is:

- the branch criterion of Lemma 8.12 is exact;
- the severity ordering it induces matches empirical pass-rate ordering at $r = 0$ across 87 offsets with Spearman correlation 0.93–0.94 on the frozen audited corpus, robustly;
- the monitor’s coverage has a specific structural explanation, though its formal universality remains an empirical observation.

Computation scripts: `scripts/cofactor_tail_model.py`, `scripts/cofactor_tail_model_v3.py`, `scripts/robustness_audits.py`, `scripts/audit4_cross_residue.py`. Data: `data/cofactor_tail_model.json`, `data/robustness_audits.json`, `data/audit4_cross_residue.json`.

8.2 Rigorous-elimination layer and the survivor corridor

The effective-budget hierarchy of Lemma 8.12 admits a natural deterministic/heuristic split at $B^{\text{eff}} = 1$.

Proposition 8.32 (Automatic-failure layer is noncovering). *Let $\mathcal{E}_r^{\leq 1} := \bigcup_{k \leq 100, \lambda: B_{r,k,\lambda}^{\text{eff}} \leq 1} \mathcal{C}_{r,k,\lambda}$ denote the union of all atlas branch-constraint classes at which the effective budget forces an automatic Erdős failure. Then for every hard residue $r \in \{0, 1716, 4862, 16302, 17160\}$, the set $\mathcal{E}_r^{\leq 1}$ has positive complement in $\mathbb{Z}$.*

Proof. For $r \in \{0, 4862, 16302, 17160\}$, the measure-sum bound

$$\mathbb{P}(t \in \mathcal{E}_r^{\leq 1}) \leq \sum_{(k, \lambda): B_{r, k, \lambda}^{\text{eff}} \leq 1} \frac{|\mathcal{C}_{r, k, \lambda}|}{\text{cm}_{r, k, \lambda}}$$

evaluates to $\{0.17, 0.45, 0.18, 0.46\}$ respectively (computed over all (k, λ) with $B^{\text{eff}} \leq 1$), all strictly less than 1. Pigeonhole then gives a positive-density complement.

For $r = 1716$, the measure-sum bound is $1.14 > 1$, so the pigeonhole argument does not directly apply. We instead exhibit explicit uncovered witnesses. The 50 values of t in `data/b_le1_uncovered_certificates.json` at $r = 1716$ (e.g. $t = 314,985,525$, giving $N = 14,548,866,415,941$; $t = 495,450,098$, giving $N = 22,884,344,578,238$; $t = 422,888,998$, giving $N = 19,532,819,930,338$) each have minimum $B_{1716, k, \lambda(t)}^{\text{eff}} \geq 2$ across all $k \in [1, 100] \setminus K_{\text{base-enriched}}(1716)$, verified independently by `scripts/verify_r1716_uncovered_witness.py`. Each such witness certifies that its arithmetic progression $\{N \equiv 1716 \pmod{M} : t \geq t_0\}$ lies outside $\mathcal{E}_{1716}^{\leq 1}$. $\square$

Remark 8.33 (Quantitative escape rates). Monte Carlo estimates of $\mathbb{P}(t \in \mathcal{E}_r^{\leq 1})$ at $N = 10^5$ random $t \in [0, L_r)$, where L_r is the lcm of branch moduli, give per-residue coverage fractions in the range 10–38% (see `scripts/covering_analysis.py`). The rigorous automatic-failure layer does not by itself imply finiteness of Erdős candidates at any hard residue; for that, the sparse-shape layer $B \in \{2, 3\}$ is required (Proposition 8.36 and Remark 8.34).

Remark 8.34 (Survivor corridor and the $B \leq 3$ sparse-shape layer). The Erdős candidate set at a hard residue r survives $\mathcal{E}_r^{\leq 1}$ with probability 10%–90% (Proposition 8.32); the survivors are further constrained by the next tightness layer $B^{\text{eff}} \in \{2, 3\}$, each of which forces the primitive cofactor $Z_{r, k, \lambda}(t)$ to lie in $\{1, p, p^2\}$ (or, after Proposition 8.36, in $\{1, p\}$ at $k \leq 100$). Conditioning further on actual-primality of these shape-forced Z ’s, Monte Carlo at $N = 3,000$ random $t \in [10^8, 10^9]$ per residue gives “E3Q survivor” rates of 2.00%, 0.17%, 0.33%, 0.53%, 0.10% for $r \in \{0, 1716, 4862, 16302, 17160\}$, reductions of 50–1600 $\times$ from the $B^{\text{eff}} \leq 1$ rates. The distribution of forced-prime constraint counts among E3Q survivors per residue is:

r	0	1	2	3	4	5	total
0	19	19	19	2	1	0	60
1716	0	0	0	4	0	1	5
4862	0	3	5	2	0	0	10
16302	0	6	5	2	2	1	16
17160	0	0	0	2	1	0	3

At $r = 0$, 31.7% of E3Q survivors carry *zero* forced-prime $B \leq 3$ constraints: they land on $B^{\text{eff}} \geq 4$ branches at every omitted k and bypass the forced-prime layer entirely. These “loose corridors” dominate the residual E3Q density at $r = 0$ and $r \in \{4862, 16302\}$, and must be modeled separately from the tight corridors at $r \in \{1716, 17160\}$ where every E3Q survivor carries ≥ 3 forced-prime constraints. See `scripts/e3q_forced_prime_histogram.py` and `data/e3q_forced_prime_histogram.json`.

Remark 8.35 (Structural primitivity of the cofactor form). At every atlas branch (r, k, λ) , the primitive cofactor $Z_{r, k, \lambda}(u) = \alpha \cdot u + \beta$ has $\gcd(\alpha, \beta) = 1$ where the slope

$$\alpha = \prod_{p \notin S_{r, k}} p^{a_p - c_p(r, k)}$$

is a product of the non-active small primes (with $S_{r,k}$ the active-prime set and $a_p = v_p(A_{\text{const}})$, $c_p(r, k) = v_p(C_{r,k})$, and $a_p > c_p(r, k)$ for $p \notin S_{r,k}$). Empirically verified at all 1,175 branches with $B^{\text{eff}} = 2$ and all 760 branches with $B^{\text{eff}} = 3$ at $k \leq 100$, and agreeing with the structural derivation. Consequently Z admits no fixed-divisor obstruction beyond the atlas factorization itself, ruling out the simplest class of rigorous branch contradictions. See `scripts/b23_shape_contradiction_scan.py` and `data/b23_shape_contradictions.json`.

Proposition 8.36 (QNR square-kill certificate theorem at $k \leq 100$). *For every atlas branch (r, k, λ) with $r \in \{0, 1716, 4862, 16302, 17160\}$, $k \in [1, 100] \setminus K_{\text{base-enriched}}(r)$, and $B_{r,k,\lambda}^{\text{eff}} = 3$, there exists a prime q dividing $\alpha_{r,k,\lambda}$ such that $\beta_{r,k,\lambda}$ is a quadratic non-residue modulo q (or $q = 2$ with $v_2(\alpha) \geq 2$ and $\beta \not\equiv 1 \pmod{4}$). Consequently $Z_{r,k,\lambda}(u)$ is never a perfect square, and the branch shape requirement $\tau(Z) \leq 3$ collapses to “ Z prime or $Z = 1$ ”.*

Proof. Finite certificate verification. For each of the 1,242 branches (r, k, λ) with $B_{r,k,\lambda}^{\text{eff}} = 3$ at $k \leq 100$, the accompanying data file `data/b3_qnr_certificates_k100.json` records $(\alpha, \beta, q, \beta \bmod q, (\beta/q))$ with either (i) q an odd prime, $q \mid \alpha$, $(\beta/q) = -1$, or (ii) $q = 2$, $v_2(\alpha) \geq 2$, $\beta \bmod 4 \neq 1$ (or, when $v_2(\alpha) \geq 3$, $\beta \bmod 8 \neq 1$). In case (i), $Z(u) \equiv \beta \pmod{q}$ is a quadratic non-residue for all u , so Z is not a perfect square. In case (ii), $p^2 \equiv 1 \pmod{4}$ (resp. $\pmod{8}$) for all odd primes p , so $Z \neq p^2$. Since $\gcd(\alpha, \beta) = 1$ by Remark 8.35, $q \nmid \beta$, closing the $p = q$ loophole. Exhaustiveness (saturation lemma). If $B_{r,k,\lambda}^{\text{eff}} = 3$ then $(k+2)/\tau(A_{r,k,\lambda}) \geq 3$, so $\tau(A_{r,k,\lambda}) \leq (k+2)/3 \leq 34$ for $k \leq 100$. Each active prime $p \in S$ contributes a factor $a_p + \lambda_p + 1 \geq 2$ to $\tau(A_{r,k,\lambda})$, so $a_p + \lambda_p + 1 \leq 34$ for every $p \in S$, forcing $\lambda_p \leq 32$. The enumeration of $B = 3$ branches at cutoff $\lambda \leq 50$ is therefore rigorously saturated, and indeed the branch count agrees at cutoffs $\lambda \leq 30$ and $\lambda \leq 50$ (both give 1,242). See `scripts/b3_qnr_certificate_k100.py` and `scripts/verify_b3_qnr_certificates_k100.py`. $\square$

Remark 8.37 (Extension at $k \leq 1000$). The analogous scan at $k \leq 1000$ produces QNR-certificate coverage $\{100.00, 98.20, 98.83, 98.79, 97.77\}\%$ for $r \in \{0, 1716, 4862, 16302, 17160\}$; overall 98.70% of 12,495 branches. The 1.30% residual is consistent with the $2^{-\omega(\alpha)}$ random-QR exception rate given the typical number of distinct prime factors of α . The extension is therefore an empirical density regularity, not a finite-certificate theorem. Scripts and data: `scripts/b23_qnr_deep.py`, `data/b23_qnr_deep.json`.

Remark 8.38 (Interpretation: where a deterministic proof can still live). Remark 8.34, Proposition 8.36 and Remark 8.37 together describe the current proof frontier. The $B^{\text{eff}} \leq 1$ layer is deterministic but non-covering. The $B^{\text{eff}} = 2, 3$ layer is uniformly shape-collapsed to “ Z prime or $Z = 1$ ” at $k \leq 100$ (by Proposition 8.36), and heuristically collapsed at 98.70% of $B^{\text{eff}} = 3$ branches at $k \leq 1000$; but the collapsed layer does not yield a rigorous per-branch contradiction (since “ Z prime” is not structurally excluded). A deterministic-finiteness route from this framework would need to combine (i) pairwise gcd-at- t constraints on simultaneous prime cofactors, (ii) covering-system arguments on the linear forms $Z_{r,k,\lambda}$, or (iii) shape obstructions beyond QNR (cubic/quartic characters). The current paper does not pursue these; conditional thinning statements follow from Proposition 8.14 combined with Bateman–Horn on the resulting shape conditions.

9 Universal admissibility

Theorem 9.1 (Universal admissibility of the core sub-tuple, Lean-verified). *For each $r \in \{0, 1716, 4862, 16302, 17160\}$, the 13-form core sub-tuple `hardRootCoeffs` is admissible*

at every prime p .

Proof. For $p > 13$: each of the 13 forms contributes at most one forbidden residue mod p ; pigeonhole gives a surviving residue. For $p \leq 13$: discharged by `native_decide` independently at each hard residue (each generating its own computational verification axiom; no sharing between residues). $\square$

Remark 9.2 (Full-tuple admissibility, Lean-verified). The full budget- ≤ 3 tuples, of sizes 13, 17, 15, 15, 17 respectively, are admissible at every prime p . Small primes (p up to the tuple length) are discharged by `native_decide` per residue; larger primes follow by the same pigeonhole bound used in the core-sub-tuple proof (`numForbiddenAtFull_le_length` + a per-residue length equality). See `hard_root_r_{0,1716,4862,16302,17160}_full_admissible_all` in `Erdos647HardRootAdmissibility.lean`. The underlying coefficient data is in `data/hard_root_forms_r*.json` and `data/admissibility_new_hard_residues.json`.

10 Barrier theorem

Theorem 10.1 (Barrier theorem, Lean-verified including CRT). *For any budgeted affine state with m active clauses and any finite set S of regular primes each greater than m , there exists a residue avoiding all clause divisibilities simultaneously.*

Proof. Induction on $|S|$ via `Finset.induction`. At each step, `one_prime_escape` gives a good residue at the new prime; CRT-composition (proved via `ZMod` arithmetic) combines with the inductive hypothesis. $\square$

11 Conditional prime-core resolution

Theorem 11.1 (Conditional, under Dickson). *Under Dickson’s conjecture for each of the 5 specific admissible tuples, the budget- ≤ 3 prime core is infinitely often solvable.*

The passage from “prime core solvable” to “full #647 witness” requires controlling the infinite family of budget- ≥ 4 constraints. Section 8 records the empirical content at the finite end of this family: for every strong-threshold record observed to date (at any residue), some $k \in \{14, 15, 16\} \cup \{7, 11, 13\}$ witnesses an Erdős-inequality failure, and the minimum killer set $S_0 = \{14, 15, 16\}$ covers all 228 strong-threshold records at $r = 0$. However, Remark 8.9 shows that the three finite-incompatibility programs we tested fail to produce a proof of this phenomenon; the natural next step is a density-level control on the budget- ≥ 4 Dickson-set tail, with an analytic bound on the tuple-restricted Titchmarsh-divisor problem on the Dickson set as the nearest proxy target. A proxy bound of this form would be a major step toward closing the Dickson-to-#647 gap, but a separate bridge from the divisor proxy to the density-level statement remains open (direct Markov/Chebyshev on fixed-threshold divisor moments does not suffice). We do not establish either bound here and flag both as principal open questions for future work.

We also record, for context, a separate Dirichlet- L pair-correlation / Stage-3B route pursued in parallel project notes. That route is not an input to the Dickson-based conditional statement of this section. In its current form it gives a conditional asymptotic theorem under GRH for primitive $L(s, \chi)$ with $q \mid \alpha$, together with a χ -twisted Hardy–Littlewood hypothesis H_χ , and a finite- T^* numerical corollary under an additional bookkeeping gate B_χ . The strict GRH-only finite- T^* individual- χ

version is stronger, and the completed recon treats it as effectively red in the present framework after two explicit failed subroutes.

A positive Dickson-conditional resolution of the hard-tuple prime core would stand in tension with the `variants.lim` intuition of Erdős (Section 1).

Proposition 11.2 (Dickson–#647 dichotomy, localized). *Assume Dickson’s conjecture holds for at least one admissible hard-tuple tuple at $r \in \mathcal{R}^* := \{0, 1716, 4862, 16302, 17160\}$. Then for some such r there exist infinitely many $t \in \mathbb{N}$ with every budget- ≤ 3 prime form at r simultaneously prime; call such t a Dickson solution at r . Exactly one of the following holds along any such sequence of Dickson solutions:*

- (A) *All but finitely many Dickson solutions t at r fail some budget- ≥ 4 constraint at $n = 2520(46189t + r)$. Consequently #647 has at most finitely many solutions.*
- (B) *Infinitely many Dickson solutions t at r satisfy all budget constraints simultaneously at $n = 2520(46189t + r)$. Equivalently, #647 admits infinitely many solutions.*

The dichotomy is a statement about the cardinality of #647’s solution set. Two structural consequences deserve separate emphasis. (i) *Relation to `variants.lim`*. The Erdős `variants.lim` strengthening asserts $\max_{m < n} (\tau(m) + m - n) \rightarrow \infty$. Case (B) directly contradicts `variants.lim` (infinitely many #647 solutions witness a bounded supremum at arbitrarily large n). Case (A) is *consistent with but strictly weaker than `variants.lim`*: finitely many solutions imply the bound is achieved only finitely often, but `variants.lim` further requires unbounded growth, which (A) alone does not guarantee. (ii) *Localization*. The reduction does not settle which of (A), (B) holds; but it pins the unresolved content to a precise arithmetic phenomenon — the budget- ≥ 4 behaviour along Dickson solutions at the 5 specific hard-tuple tuples — that a sieve-theoretic or analytic result could in principle address. In particular, the reduction is not merely a partial resolution: it localizes the *entire* incompatibility between Dickson-type conjectures and Erdős’s `variants.lim` intuition to these 5 specific arithmetic witnesses. (iii) *Target hypothesis*. A natural paper-primary analytic target is a density-level control on the budget- ≥ 4 tail along the Dickson set. The nearest literature-shaped proxy target is a *tuple-uniform Titchmarsh-type* control on the Dickson set: an upper bound for divisor moments on a tail form L_{r,k_0} averaged over the simultaneous-prime-tuple set $D_r(X) = \{t : L_{r,k} \text{ prime for all core } k\}$. Existing project notes freeze that proxy as a uniform upper bound of the shape

$$\sum_{t \leq X, t \in D_r(X)} \tau(L_{r,k_0}(t)) \ll |D_r(X)| \log X$$

for k_0 in the finite tail family. Remark 11.3 records the current unconditional state of the art on this proxy and isolates the two remaining conjectural inputs. The proxy alone would still not settle the full Track 2 tail statement, because fixed- threshold Markov/Chebyshev on moments of $\tau(L_{r,k_0}(t))$ does not directly upper-bound the good-set density on the Dickson set. At the reference residue $r = 0$, the bridge now closes conditionally as recorded in Remark 11.3; for the other four hard residues and for residue-uniform versions, the direct tail-density statement and this tuple-restricted divisor proxy remain open analytic targets. A natural conceptual framework for the proxy bound would be a prime-core-relative analog of Matthiesen’s correlation estimates for multiplicative functions on affine-linear systems (Matthiesen, *Correlations of the divisor function*, Proc. London Math. Soc. (3) 104 (2012), 827–858; *Linear correlations of multiplicative functions*, Proc. London Math. Soc. (3)

121 (2020), 372–425), controlling divisor moments on the 13-form Dickson set rather than on the ambient integers.

Remark 11.3 (Absolute-form tuple-restricted τ bound at $r = 0$ (unconditional); conditional outlook). At the reference hardest residue $r = 0$ (tuple rank 13), the proxy bound of (iii) has a clean *unconditional* form; the bridge from the proxy to a Dickson-set density-level statement sits inside the same Dickson/Schinzel conjectural frame used for the dichotomy in Proposition 11.2 and is flagged here as a *conditional outlook*, not as a result of this paper. Analogous statements at the other four hard residues $r \in \{1716, 4862, 16302, 17160\}$ (ranks 15 or 17) are expected to hold by the same composite but require residue-specific bookkeeping not executed here; we state this Remark only at $r = 0$.

(a) *Unconditional absolute form at $r = 0$.* For every tail offset k_0 in the finite budget- ≥ 4 family,

$$\sum_{t \leq X, t \in D_0(X)} \tau(L_{0,k_0}(t)) \ll_{k_0} X (\log X)^{-12}$$

unconditionally. The proof is a composition of known published tools at rank 13: a Selberg Λ^2 upper-bound sieve on the 13-form Dickson indicator $\mathbf{1}_{D_0(X)}$ (in the Maynard 2015 normalization; the Λ^2 weight majorizes $\mathbf{1}_{D_0(X)}$ pointwise on W -tricked residues at admissibility, which is verified for $r = 0$ at $p \leq 19$ via the factorization $46189 = 11 \cdot 13 \cdot 17 \cdot 19$ and the divisibility $7 \mid d_k$ for every d_k in the $r = 0$ divisor set); CRT reduction of the joint divisibility constraints to a single arithmetic progression via the W -trick (Maynard 2015 equations (5.1)–(5.2)), with the cross-form resultant prime support $\{2, 3, 5, 7, 11, 13, 17, 19, 23\}$ absorbed at $w(N) \geq 23$; and Shiu/Nair–Tenenbaum Brun–Titchmarsh for τ on a linear form in an arithmetic progression (Shiu, *A Brun–Titchmarsh theorem for multiplicative functions*, J. reine angew. Math. 313 (1980), 161–170; Nair and Tenenbaum, *Short sums of certain arithmetic functions*, Acta Math. 180 (1998), 119–144; Henriot, *Nair–Tenenbaum bounds uniform with respect to the discriminant*, Math. Proc. Camb. Phil. Soc. 152 (2012), 405–424). The rank-13 sieve integral is tabulated in the Polymath8b range (*Variants of the Selberg sieve, and bounded intervals containing many primes*, Res. Math. Sci. 1 (2014)).

(b) *Conditional outlook: Dickson-set density at $r = 0$.* Converting the unconditional absolute bound in (a) into a statement about the density of the τ -exceptional set *inside* the rank-13 Dickson set $D_0(X)$ requires a quantitative lower bound of the shape $|D_0(X)| \gg X(\log X)^{-13}$. This is the quantitative form of the rank-13 case of Dickson’s conjecture (equivalently, a specialisation of Schinzel’s Hypothesis H at the $r = 0$ 13-tuple), and it sits within the same named-conjecture framework already invoked by Erdős himself as the conditional premise under which the infinitude-variant adjacent to #647 would follow [1]. Proposition 11.2 already conditions the Track-2 dichotomy on a qualitative form of Dickson (existence of infinitely many $t \in D_0(X)$); the quantitative lower bound above is a strictly stronger hypothesis in the same conjectural frame.

Under this quantitative Dickson-type hypothesis, a Dickson-set Turán–Kubilius concentration bound for $\omega(L_{0,k_0}(t))$ on $D_0(X)$ — specifically, a second-moment Selberg variance estimate $\sum_{t \in D_0(X)} (\omega(L_{0,k_0}(t)) - \log \log X)^2 \ll |D_0(X)| \log \log X$ — implies by Chebyshev that $\{t \in D_0(X) : \omega(L_{0,k_0}(t)) \leq C\}$ has Dickson-set density $o(1)$ for every fixed C , hence the $r = 0$ Track-2 conclusion $E_0(X)/|D_0(X)| = o(1)$. This is a conditional corollary of the reduction, not a theorem of the present paper; its status and the conditional framework are recorded here to make the analytic content explicit and to distinguish it from the unconditional proxy in (a).

12 Quantitative predictions and witness search

The Bateman–Horn singular series for the 13-tuple at $r = 0$ is $C_{\text{BH}} \approx 2.3179 \times 10^8$ (primes $\leq 500,000$; convergence ratio 0.999949).

Using the leading-order Bateman–Horn asymptotic, the expected witness count is

$$\#\{t \leq X : \text{all 13 prime}\} \approx \frac{C_{\text{BH}} \cdot X}{\prod_{i=1}^{13} \log(\alpha_i X)}.$$

At $X = 10^{13}$: ≈ 0.48 expected witnesses. At $X = 10^{14}$: ≈ 2.5 expected witnesses. The figures reported are the leading-order Bateman–Horn prediction assuming full independence across the 13 forms; refinements that account for residue-dependent correlation among the forms (specifically, the common divisibility structure induced by $d_i \mid 2520$) typically reduce the leading-order prediction by 30–40% in comparable settings, but we do not attempt a full correlation-corrected prediction here. Recomputable from `scripts/paper_numerical_audit.py`.

A computational search is in progress at $r = 0$. Phase 1 ($t \leq 10^{13}$) and two parallel Phase 2 ranges partitioning $[10^{13}, 10^{14}]$ run concurrently; interim results are synced to the public repository and refreshed regularly.

Near-miss distribution (snapshot 2026-04-17). At this snapshot (Phase 1 at $\sim 17\%$; Phase 2 ranges at $\sim 2\%$ each), the combined search has observed 9 distinct 12/13 near-misses, 129 at 11/13, and 1,355 at 10/13; no 13/13 witness has been found. The 9 near-misses at 12/13 span $t = 8,618,618,190$ (the smallest, with composite form $d = 210$, composite factor 1109, above sieve limit) up to $t = 55,386,505,208,280$.

Across the full corpus of 40,984 composite form-values observed in the 10,526 near-miss records (at $k \in \{9, 10, 11, 12\}$), the per-form composite rate is essentially uniform: 19.6% to 20.8% across the 13 forms, with no form significantly over- or under-represented. At the extreme 12/13 tail ($N = 9$), the composite is distributed as 2 hits each at $d \in \{1260, 2520\}$ (the two largest) and 0 hits at $d \in \{105, 140\}$ (the two smallest); we flag this as a preliminary observation pending larger N from continued search, noting that a 4/9 concentration in the two largest of thirteen forms is not statistically distinguishable from the uniform null at this sample size. The broader uniformity across 40,984 form-values is the primary observation; the $N = 9$ tail pattern is reported as empirical data but not as a structural claim.

Under naive Bateman–Horn the expected number of 13/13 witnesses through $X = 10^{14}$ is ≈ 2.5 . The current corpus reaches $X \approx 1.01 \times 10^{14}$ and contains exactly one 13/13 hit (at $t = 11,741,230,040,310$); see Section 8 and Remark 8.9 for the empirical context. The observed-to-predicted ratio at that horizon is ≈ 0.62 , consistent with BH leading-order density to within a factor of two. Results are recomputable from `data/all_phases_merged.jsonl` and `docs/near_miss_snapshot_20260417.md`; the full quantitative refresh is in `docs/d2_bh_validation_result.md`.

Proposition 12.1 (Per-residue Bateman–Horn predictions). *The leading-order Bateman–Horn predictions across the 5 hard residues differ substantially due to tuple-size variation:*

r	K	C_{BH}	pred. at 10^{13}	pred. at 10^{14}	pred. at 10^{16}
0	13	2.32×10^8	0.47	2.54	78.68
1716	17	6.83×10^{10}	2×10^{-5}	1.4×10^{-4}	3.0×10^{-3}
4862	15	2.53×10^9	2.6×10^{-3}	1.2×10^{-2}	3.2×10^{-1}
16302	15	5.42×10^9	5.1×10^{-3}	2.5×10^{-2}	6.4×10^{-1}
17160	17	1.15×10^{11}	4×10^{-5}	2.2×10^{-4}	5.1×10^{-3}

Computed at prime bound 10^6 via `scripts/compute_bh_all_residues.py`; full data in `data/bh_all_residues.json`.

Proposition 12.2 (Case-A screen density: pointwise pass probability and cumulative expected count). *In this proposition, case-A refers to the fixed-offset truncated screen $\tau(2520N - k) \leq k + 2$ at every $k \in [1, 100] \setminus K_{\text{base-enriched}}(r)$. Any true Erdős #647 counterexample on residue r must satisfy this screen as a necessary condition (the full problem demands $\tau(n - k) \leq k + 2$ for all $1 \leq k < n$, a condition with k growing with n). We do not claim that the truncated screen alone implies #647-finiteness; it is the finite-horizon instrument used to bound what modeling can currently verify. The offsets $k > 100$ and the moving-tail regime $k \rightarrow \infty$ are not treated in this proposition. A moving- K window scan (see Remark 8.17 and `scripts/moving_k_window_experiment.py`) shows that for $k \in [101, 1000]$, per- k empirical pass rates are ≥ 0.50 at every residue with means converging to 1 as K grows (mean 0.96 at $[101, 200]$, 0.99 at $[201, 500]$, 0.999 at $[501, 1000]$). No very-tight offsets (< 0.3) appear at $k > 100$; the tight pressure is concentrated at $k \leq 100$.*

We separate two distinct quantities that were conflated in an earlier draft.

(A) Pointwise omitted-offset pass probability $p_r(t)$. The probability that $\tau(2520N - k) \leq k + 2$ holds at every $k \in [1, 100] \setminus K_{\text{base-enriched}}(r)$ for t sampled uniformly at scale t . Directly measured at $t \sim 10^{8.5}$ via per- k -independence product (calibration (ii) below), extrapolated to other scales via per- B^{eff} pass-rate drift observed in the scale-stability experiment. This quantity decreases with t (primes and low- τ integers become rarer as affine forms grow).

(B) Expected cumulative case-A count $E_r([T_0, X])$.

$$E_r([T_0, X]) \approx \int_{T_0}^X \rho_r(t) p_r(t) dt,$$

where $\rho_r(t) = C_{\text{BH}}/(\log(\alpha \cdot t))^{m_r}$ is the Bateman–Horn base-tuple hit-density at scale t and T_0 is the lower integration limit. We report results with $T_0 = 10^8$ (the floor of the scale-stability calibration range); integrating from $T_0 = 2$ gives values 10^2 – 10^6 times larger per residue but relies on the scale model extrapolated below its fitted range, so we do not treat it as load-bearing. The horizon X is the t -horizon throughout this proposition ($X = X_t$), not the N - or n -horizon. This quantity is non-decreasing in X .

Two active calibrations plus a legacy heuristic. Each is computational or heuristic; none is a proved density theorem. Calibrations (i) and (ii) are active; (iii) is a retired historical placeholder retained only for transparency.

(i.a) Raw random- t empirical upper bound. A case-A pass scan at $t \in [10^8, 10^9]$ with $n = 1,000$ samples per residue (5,000 pooled) gives 0/1000 per residue, 0/5000 pooled. One-sided 95% Clopper–Pearson on the raw IID sample: $P(\text{full pass}) \leq 1 - 0.05^{1/1000} \approx 3.0 \times 10^{-3}$ per residue, pooled $\leq 1 - 0.05^{1/5000} \approx 6.0 \times 10^{-4}$. This is a statistical confidence bound on the raw- t distribution at $t \sim 10^{8.5}$.

- (i.b) Recorded near-miss corpus diagnostic. *The full $r = 0$ witness-search corpus recorded to $t \sim 10^{14}$ contains 26,253 logged near-miss records (all primes ≥ 9 enriched forms). All 26,253 fail case-A; in particular, the single 13/13 shape-pass (the confirmed witness at $t = 11,741,230,040,310$) fails case-A at $k = 7$ because $\tau(2520N - 7) = 16 > 9 = k + 2$ (direct computation). Interpreting the corpus as a sample from the recorded-near-miss distribution, one-sided 95% Clopper–Pearson upper bounds per stratum are:*

stratum	sample n	UB on $P(\text{full pass} \mid \text{stratum})$
$C_{\geq 9}$	26,253	1.1×10^{-4}
$C_{\geq 11}$	330	9.0×10^{-3}
$C_{\geq 12}$	21	0.133
$C_{\geq 13}$	1	0.951

The $C_{\geq 13}$ row is a single observation and is diagnostic rather than estimatory: the sole 13-tuple shape-pass fails case-A at $k = 7$, falsifying the implicit assumption that shape-pass implies near-full-pass, without yielding a useful conditional probability estimate. The corpus-wide $C_{\geq 9}$ bound is a statistical statement about the recorded-near-miss layer, not the raw- t distribution; combining with (i.a) for a full per- t bound requires an explicit model of the recorded layer’s density within the search range.

- (ii) Atlas-derived independence-product estimate (pointwise). *From $n = 2,000$ -sample per- k pass rates at $t \in [10^8, 10^9]$, the per- k -independence product evaluates to: 3.11×10^{-10} ($r = 0$), 1.66×10^{-10} ($r = 1716$), 1.16×10^{-10} ($r = 4862$), 1.92×10^{-10} ($r = 16302$), 1.60×10^{-10} ($r = 17160$). These are values of p_r at $t \sim 10^{8.5}$. The scale-stability experiment at three t -scales spanning four orders of magnitude fits $p_r(t) \approx p_r(10^{8.5}) \cdot 10^{\gamma_r \cdot (\log_{10} t - 8.5)}$ with per-residue slopes $\gamma_r \in [-3.4, -2.0]$.*
- (iii) Legacy witness-scale heuristic (not a measurement). *Earlier drafts used $P(\text{full case-A pass} \mid \text{shape hit}) \approx 10^{-14}$ as a calibration anchor, rationalized by dividing the one confirmed shape hit by the search range $\sim 10^{14}$. The per- t verification in (i.b) shows that shape-pass does not imply near-full-pass: the witness fails at $k = 7$. The 10^{-14} figure is therefore a legacy order-of-magnitude placeholder, not a direct measurement; we retain it only as historical context and do not use it in the quantitative synthesis below. Combined with $\rho_r(t)$, yields a cumulative estimate $E_r(X) \approx 10^{-14} \cdot N_{\text{BH},r}(X)$ where $N_{\text{BH},r}$ is the expected shape-hit count.*

Table of pointwise probabilities $p_r(t)$ (calibration (ii) with scale extrapolation):

$\log_{10} t$	$r = 0$	$r = 1716$	$r = 4862$	$r = 16302$	$r = 17160$
8.5	3.1×10^{-10}	1.7×10^{-10}	1.2×10^{-10}	1.9×10^{-10}	1.6×10^{-10}
10.5	5.4×10^{-12}	2.0×10^{-13}	1.7×10^{-13}	8.8×10^{-13}	1.8×10^{-13}
12.5	9.3×10^{-14}	2.5×10^{-16}	2.5×10^{-16}	4.1×10^{-15}	2.1×10^{-16}
14.5	1.6×10^{-15}	3.1×10^{-19}	3.8×10^{-19}	1.9×10^{-17}	2.3×10^{-19}

Table of cumulative expected counts $E_r([10^8, X])$ (integral density model, calibration (ii); integration floor $T_0 = 10^8$ at the scale-stability calibration lower bound):

X	$r = 0$	$r = 1716$	$r = 4862$	$r = 16302$	$r = 17160$
10^{14}	1.2×10^{-13}	2.3×10^{-17}	6.0×10^{-16}	1.1×10^{-15}	2.2×10^{-17}
10^{17}	1.3×10^{-13}	2.3×10^{-17}	6.0×10^{-16}	1.1×10^{-15}	2.2×10^{-17}
10^{19}	1.3×10^{-13}	2.3×10^{-17}	6.0×10^{-16}	1.1×10^{-15}	2.2×10^{-17}

Most residues saturate quickly within the calibration range because $p_r(t)$ decays faster than $\rho_r(t)$ grows (per-residue slope $\gamma_r \in [-3.4, -2.0]$). At $r = 0$ the slope $\gamma_0 = -2.03$ is shallowest and $E_0([10^8, X])$ grows modestly from 1.0×10^{-13} at $X = 10^{12}$ to 1.3×10^{-13} at $X = 10^{19}$. This saturation is the sharpest quantitative consequence of the density model: for $r \in \{1716, 4862, 16302, 17160\}$ the integrated expected counterexample count $E_r([10^8, X])$ is essentially constant for all $X \geq 10^{13}$, and for $r = 0$ it grows by at most $\sim 30\%$ between $X = 10^{12}$ and $X = 10^{19}$. Under the current calibration the integrated case-(A) counterexample risk is therefore approximately scale-independent at large X_t , with total expected count bounded above by 1.3×10^{-13} (or 8.1×10^{-13} under the anchored log-law) at $r = 0$ and by smaller values at the other four residues, across all X_t in the modeled range.

Sensitivity envelope (integral under $\gamma_r \pm 0.5$) at $X = 10^{19}$: for $r = 0$ with $\gamma_0 = -2.03$, $E_0([10^8, 10^{19}])$ evaluates to 1.0×10^{-12} at $\gamma + 0.5$ (less-negative slope, slower decay, larger tail integral), 1.3×10^{-13} at the fitted value, and 7.6×10^{-14} at $\gamma - 0.5$ (more-negative slope, faster decay, smaller integral). Total range factor ~ 14 ; other residues vary by $< 3\times$ under the same envelope.

Calibration-floor caveat. Integrating from $T_0 = 2$ rather than $T_0 = 10^8$ yields values 10^2 – 10^6 times larger per residue, but relies on the scale model extrapolated below its fitted range and is not load-bearing. The $[T_0, X]$ framing is the credible form.

Comparison with log-law tail model and witness-calibrated (iii). Calibration family (ii) has two variants: (ii.a) the local power-law fit $p_r(t) \propto t^{\gamma_r}$, and (ii.b) an anchored Landau-Sathe log-law tail $F_B(x) \propto (\log \log x)^{s-1}/(\log x)$. The Landau model is calibrated from the scale-stability data for the decay shape and anchor-normalized at $t_0 = 10^{8.5}$ to the empirical independence product (correcting a factor- ~ 6 – 653 per-residue normalization offset in the raw Landau fit). At $X = 10^{19}$ the cumulative estimates are:

r	(ii.a) power-law	(ii.b) anchored log-law	(iii) witness-calibrated
0	1.3×10^{-13}	8.1×10^{-13}	3.3×10^{-9}
1716	2.3×10^{-17}	2.7×10^{-16}	1.7×10^{-12}
4862	6.0×10^{-16}	7.7×10^{-15}	7.6×10^{-11}
16302	1.1×10^{-15}	1.1×10^{-14}	7.6×10^{-11}
17160	2.2×10^{-17}	2.8×10^{-16}	1.7×10^{-12}

The anchored log-law (ii.b) gives values 6–13 $\times$ larger than the power-law (ii.a) per residue, consistent with the Landau asymptotic decaying more slowly than a local power-law fit. Both variants of family (ii) predict negligible expected counts at the modeled horizons, and are consistent with the empirical upper bounds of (i.a) and (i.b). The legacy heuristic (iii) gives larger values (10^3 – $10^4 \times$ compared with (ii)), but the witness verification in (i.b) shows (iii) was never a direct measurement; its numerical value is retained only for historical comparison. See `scripts/loglaw_tail_model.py` and `data/loglaw_tail_model.json`.

Modeling caveats. Four specific sources contribute to the gap between (ii) and (iii): (a) the independence-product assumption in (ii) (cross- k correlations may change (ii) by several orders of magnitude in either direction; for omitted offsets $n - a, n - b$ with primes $q > |a - b|$ the two divisibility indicators cannot simultaneously vanish, giving a negative covariance at each such prime, while local

Bateman–Horn factors can contribute positively — the sign and size of the net effect on joint low- τ pass events is not yet modeled); (b) the marginal-over- t versus per-shape-hit framework mismatch; (c) the single-data-point variance in (iii); and (d) the power-law scale extrapolation $p_r(t) \propto t^{\gamma_r}$ used in (ii). Regarding (d): per- B^{eff} pass rates are asymptotically governed by Landau / PNT-type log-law probabilities $F_b(Z) \propto (\log \log Z)^{b-2}/(\log Z)$, not by power laws in t . The anchored log-law variant (ii.b) above uses the Landau-Sathe decay shape anchored to the empirical independence product at $t = 10^{8.5}$. At $X = 10^{19}$ the anchored log-law gives values $6\text{--}13\times$ larger than the power-law, consistent with a slower-decaying asymptotic. This is a modest correction relative to the overall $10^3\text{--}10^4\times$ gap between family (ii) and the witness heuristic (iii): the tail-law choice is not a first-order source of the remaining uncertainty. We consider caveat (d) substantially tested and demoted.

Regarding (a): we tested the independence-product assumption by computing direct block ratios $R_S = \Pr(\bigcap_{k \in S} \text{Pass}_k) / \prod_{k \in S} \Pr(\text{Pass}_k)$ for four block families at each hard residue: the monitor $\{7, 11, 13, 16\}$, the empirical minimum cover $\{15, 16, 27\}$, the top-5 bottleneck offsets from calibration (ii.a), and the union of monitor and top-5 bottlenecks. At $n = 5,000$ samples per residue and $t \in [10^8, 10^9]$, all twenty R_S values fall in $[0.63, 1.80]$, with $\max |\log R_S| \approx 0.59$. Independence holds empirically to within factor ~ 2 at the block level. The $10^3\text{--}10^4\times$ gap between (ii) and (iii) is therefore not attributable to cross- k dependence; caveat (a) is substantially tested and demoted.

The remaining primary source of modeling uncertainty is caveat (b), the marginal-over-raw- t versus per-shape-hit conditioning mismatch. Calibration (ii) samples t uniformly while calibration (iii) is conditional on a base-tuple prime hit. A weighted base-tuple conditional surrogate is the next natural refinement. Neither calibration is a proved bound; all are extrapolated models with documented uncertainties.

See `scripts/density_integral_model.py`, `data/density_integral_model.json`, `scripts/scale_stability_test.py`, and `data/scale_stability_test.json`.

The largest single- k bottleneck in (ii) is $k = 16$ (per- k pass rate ≈ 0.35 at every residue), consistent with Proposition 8.16. See `scripts/case_a_density_factored.py` and `data/case_a_density_factored.json` for the full per- k rate tables; `docs/case_a_density_model.md` for caveats.

Remark 12.3 (Tuple-size convention and divisor-extension sizes). Proposition 12.1 uses the hard-root tuple sizes $K \in \{13, 17, 15, 15, 17\}$ at which the singular series were originally computed. The divisor-extension tuple sizes from Section 6 and Theorem 6.5 are $\{13, 15, 14, 14, 15\}$ at $r \in \{0, 1716, 4862, 16302, 17160\}$. The two conventions agree at $r = 0$, so the $r=0$ entry of the table is directly comparable to the divisor-extension framework used elsewhere in the paper. At nonzero residues, the divisor-extension convention has smaller K by 1 or 2; at $X = 10^{14}$ this shifts the predicted density upward by a factor of $\log(\alpha X)^{K_{\text{old}} - K_{\text{new}}} \approx 10^{1.5}$ to 10^3 , which does not change the order-of-magnitude picture (nonzero residues remain effectively zero at current search horizons). Proper recomputation of the singular series on the divisor-extension tuples is deferred to follow-up work; for the paper’s structural conclusions (the empirical killer-set discussion of Section 8 and the conditional resolution of Section 11) the effect is within existing order-of-magnitude statements.

Remark 12.4 (Informational focus of the $r = 0$ search). The $r = 0$ residue is by far the most informative computational target. At $X = 10^{14}$, the two 17-form residues ($r \in \{1716, 17160\}$) have expected witness counts roughly 10^4 below $r = 0$, while the two 15-form residues ($r \in \{4862, 16302\}$) are roughly 10^2 below $r = 0$. The separation arises from the tuple-size factor in the denominator product $\prod \log(\alpha_i X)$: each additional form incurs a $\log X$ penalty, and at $\log 10^{14} \approx 32$, even two extra forms reduce the predicted density by a factor of $\approx 10^3$. The asymmetry is robust in its

direction to refined-formula corrections (correlation adjustments to the singular series typically move predictions by factors of a few to at most an order of magnitude, not by the four orders separating the 17-form residues from $r = 0$); we have not, however, computed the refined predictions for the non- $r = 0$ residues.

The practical consequence: a null result from the witness search at $r = 0$ (Phases 1 and 2) is informative — at $X = 10^{14}$ the probability of observing zero witnesses under leading-order Bateman–Horn is $e^{-2.5} \approx 8\%$. Null results at the other four hard residues at any feasible $X \leq 10^{14}$ are, by contrast, fully consistent with Bateman–Horn and carry little evidence against witness existence. The search strategy’s focus on $r = 0$ reflects this informational asymmetry; extending the search to $r \in \{1716, 4862, 16302, 17160\}$ would not be a productive use of computational resources in the near term.

13 The reduction chain

The Lean formalization packages the large- n conclusion as a pair of theorems. The conditional reduction

`erdos647_conditional_on_openResiduesStage1`: $\forall n, 24 < n \rightarrow \neg \text{Candidate } n$,

quantified over any hypothesis asserting that no N with $N \bmod M \in \text{openResiduesStage1}$ yields an Erdős #647 candidate, captures the entire structural reduction. The full (unquantified) theorem

`erdos647_with_stage1_axiom`: $\forall n, 24 < n \rightarrow \neg \text{Candidate } n$

applies the single intentional Stage-1 axiom `openResiduesStage1_no_solution` to the conditional form. Its current dependency profile is:

Step	Theorem	Status
Finite coverage ($25 \leq n \leq 84$)	<code>candidate_false_finite</code>	proved (native_ decide)
Candidate $\rightarrow$ IsErdos647	<code>bridge_candidate_to_isErdos647</code>	proved
2520-divisibility	<code>erdos647_div_2520</code>	proved
Sieve reduction (Bridge 2)	<code>bridge_isErdos647_to_sieve</code>	proved
55-residue closure	<code>singleOverlapResidues42</code> $\cup$ <code>directFullValueResidues13</code>	proved
Admissibility ($\times 5$)	<code>hard_root_r*_admissible_all</code>	proved
Open residues ($\times 41$)	<code>openResiduesStage1_no_solution</code>	axiom

No source-level `sorry` remains in `lean/`. The prior Bridge 3 budget-transfer lane has been retired (the peeled-cofactor implication was shown to be pointwise unsound; see `docs/bridge3_design_memo_20260421.md`) and `Bridge3.lean` is now a 16-line placeholder stub. The operative closure surface is the Stage-1 partition

`closedResiduesStage1` = `singleOverlapResidues42` $\cup$ `directFullValueResidues13`,
`openResiduesStage1` = `survivingResidues96` $\setminus$ `closedResiduesStage1`,

with `closedResiduesStage1_card = 55` and `openResiduesStage1_card = 41`, both proved by `native_decide`. Bridge 2 is closed in `Bridge2.lean`; the substantive sub-lemma `tau_ge_large_noncoprime` is proved across all six non-trivial coefficient values.

The narrow axiom `openResiduesStage1_no_solution` is confined to the 41 residues in the exact-v2 open set and represents the genuinely open content of Problem #647. It subsumes both the historical hard-14 core (five original plus nine exact-v2-reopened residues) and the twenty-seven residues whose closures under the invalidated Bridge-3 peeled-cofactor mechanism require a sound replacement. A post-retirement audit of the 55 Lean-closed residues (`docs/audits/55_closed_post_bridge3_audit_20260424.md`) confirms none of the 55 depends on the Bridge-3 transfer; the single-overlap and direct full-value arguments work directly at the full-value level $2520 \cdot N - k$ and rely only on divisor monotonicity and sound coprime-multiplicativity with explicit coprimality hypotheses.

Remark 13.1 (v_{23} Hensel-rootline uniformity of the sieve-surviving residues). The 55-closed / 41-open partition in the Stage-1 axiom reflects the structural reach of this paper’s closure mechanisms (`SingleOverlapClosure`, `DirectFullValueClosure`), not a fundamental hardness gradient between the two classes. A uniform cycle-3-style partial-closure mechanism applies to every one of the 96 sieve survivors via the prime 23: for each residue $r \in \text{survivingResidues96}$, there exist shift $k \in \{1, \dots, 95\}$ and a sub-AP $\{N \equiv b_r \pmod{23^3 \cdot M}\}$ of density $23^{-3} = 1/12167$ on which $2520 \cdot N - k = d(r, k) \cdot (\alpha \cdot s + c_0)$ algebraically with $c_0 \geq 2$ and $\tau(d \cdot p) > k + 2$ for every prime p , so that no N on the sub-AP is `good0n` at k . The structural basis is $\gcd(23, 2520) = \gcd(23, M) = 1$, so 23 is free of both the divisor budget $2520 = 2^3 \cdot 3^2 \cdot 5 \cdot 7$ and the sieve modulus $M = 46189 = 11 \cdot 13 \cdot 17 \cdot 19$; the 23-adic Hensel tower is therefore available at every residue class. An empirical scan verified positive-margin closures at 136/136 tested residues (the 41 open, the 55 closed, and 40 sampled non-surviving residues outside `survivingResidues96`). Consequently the present paper’s 55/41 split is an artifact of *which* mechanisms are formalized in the 55 closures and not a structural claim about the 41 being harder; a multi-shift primitive combining several such per-residue sub-AP closures (design in `docs/lean_design/multishift_primitive_design_20260424.md`) would discharge `openResiduesStage1_no_solution` uniformly. Per-residue certificate data are tabulated in `data/all41_v23_hensel_certificates_20260424.json`; full findings are recorded in `docs/audits/all41_hensel_v23_unified_20260424.md` and `docs/audits/v23_universal_pattern_20260424.md`.

14 Formal verification summary

File	Sorry	Lines
Erdos647Main	0 (legacy wrapper retired)	88
Erdos647SieveCertificate	0	64
Erdos647HardRootAdmissibility	0	314
Erdos647Closure92_Legacy (historical)	0	94
Erdos647BarrierTheorem	0	252
Erdos647BridgeV1	0	904
Erdos647B23Closure	0	316
Erdos647ClosureCerts	0	267
Erdos647ClosureData	0	2837
Erdos647BranchForms	0	1212
Erdos647ReductionChain	0	118
Erdos647ResiduePartitionStage1	0	124
Erdos647Stage1Axiom	0 + 1 narrow axiom	27
Erdos647ConditionalFiniteWindow	0	152
Bridge2	0	1044
Bridge3 (stub, retired)	0	16
Total (proof-critical subset)	0 sorry + 1 axiom	7829

All finite-computational claims are discharged via Lean’s `native_decide`. The proof-critical chain is conditional on the Stage-1 axiom wrapper: `Erdos647ReductionChain.erdos647_conditional_on_openResiduesStage1` states the theorem relative to the open residue set, and `Erdos647Stage1Axiom.erdos647_with_stage1_axiom` applies the single intentional axiom `openResiduesStage1_no_solution` to yield the full result. Bridge 3 is retired as a stub; no source-level `sorry` remains in `lean/`. The complete Lean development spans 49 files; the above table records the proof-critical subset. The non-definitional dependency profile is: `propext`, `Classical.choice`, `Quot.sound`, `Lean.ofReduceBool` (from `native_decide`), `openResiduesStage1_no_solution`.

Trust chain and reproducibility. All residue closures, shape decompositions, sieve enumerations, cofactor factorisations, and arithmetic verifications in this paper are discharged inside the Lean 4 kernel. Finite computational claims (residue-list cardinalities, divisibility checks, primality over fixed small ranges, enumeration of surviving residues mod M) use Lean’s `native_decide`, which compiles the relevant decidable proposition to native code and trusts the result via the `Lean.ofReduceBool` axiom listed above. No step of the verified chain depends on externally-computed data ingested from outside Lean, on unsafe Lean features (`unsafe`, `extern`), or on trusted code outside the Lean source tree. The residue-list literals `closedResiduesStage1`, `singleOverlapResidues42`, `directFullValueResidues13`, `openResiduesStage1`, and `survivingResidues96` are Lean-internal `Finset ℕ` definitions, and the partition equalities and cardinality claims are proved by `native_decide` on those definitions. The post-retirement audit at `docs/audits/55_closed_post_bridge3_audit_20260424.md` further verifies that the 55 closures do not depend on Bridge-3-adjacent reasoning: all multiplicativity uses in `Bridge2` and downstream files carry an explicit `Nat.Coprime` hypothesis, so the pointwise-unsound peeled-cofactor identity never appears in the closure proofs.

References

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- [6] S. Dutta, comments on Erdős Problem #647, <https://www.erdosproblems.com/647>: initial derivation at 13:40 on 18 Jan 2026 (modular constraints $n = 2520N$ with $d_iN - 1$ prime or almost-prime for $d_i \in \{1260, 840, 630, 420, 315, 210, 105\}$), and refined form at 17:49 on 19 Jan 2026 ($315N - 1 \in \{p, 2p\}$, $105N - 1 \in \{p, p^2, 2p, 4p\}$; $2520N - 1$ upgraded to prime after the mod-4 observation).
- [7] B. Alexeev, comment on Erdős Problem #647 at 00:50 on 19 Jan 2026, <https://www.erdosproblems.com/647>: flags coprimality gap in Dutta’s $k \in \{8, 24\}$ step, upgrades $2520N - 1$ to prime via $2520N - 1 \equiv 3 \pmod{4}$, conjectures $11 \mid n$ and $13 \mid n$, and exhibits the explicit near-witness $n = 604,517,614,941,240$ (with $N = 239,887,942,437$) for which $\tau(n - k) \leq k + 2$ holds for $1 \leq k \leq 13$ and $k = 24$.
- [8] J. Agbanwa, submission on Erdős Problem #647, <https://www.erdosproblems.com/647> forum comment at 15:18 on 28 Jan 2026, with companion Lean formalization at <https://zenodo.org/records/18390414>. Flagged as incorrect in subsequent forum review.
- [9] T. Tao, comments on Erdős Problem #647, <https://www.erdosproblems.com/647>: forum comment at 22:41 on 2 Oct 2025 (curated description update on the τ/ω hierarchy) and at 16:10 on 28 Jan 2026 (response to [8]).
- [10] T. Tao, J. Teräväinen, *Quantitative correlations and some problems on prime factors of consecutive integers*, preprint arXiv:2512.01739 (1 Dec 2025). Establishes a general two-point correlation estimate for bounded multiplicative functions with logarithmic savings (via Pilatte correlations), a Maynard-type sieve argument for $\omega(n + k) \leq \Omega(n + k) \ll k$ infinitely often (resolving Erdős #248), and an asymptotic for $\#\{n \leq x : \omega(n) = \omega(n + 1)\}$ at almost all x . Remark 1.2 of the preprint explicitly flags Erdős #647 as “beyond the capability of our methods,” noting that the small- k barrier-case variant is of “comparable difficulty to the prime tuples conjecture.”
- [11] C. F. Lau, *On the Number of Prime Factors of Consecutive Integers*, preprint arXiv:2604.15042 (16 Apr 2026). Refines [10] to $\omega(n + k) \ll \log k$ for $k \geq 2$ on an infinite sequence; cited on the curated Bloom #647 page [1].